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**Method for determination of
tetra-through octachlorodibenzo-
p-dioxins, tetra-through
octachlorodibenzofurans and
dioxin-like polychlorinatedbiphenyls
in stationary source emissions**

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Foreword

This translation has been made based on the original Japanese Industrial Standard revised by the Minister of Economy, Trade and Industry through deliberations at the Japanese Industrial Standards Committee in accordance with the Industrial Standardization Law. Consequently **JIS K 0311 : 1999** is replaced with this Standard.

In this revision, this Standard has been extensively looked over and the revision has been carried out about definitions, sampling of sample gas, pretreatment of sample, identification and determination, report of results and quality control of measurement data.

Attention is drawn to the possibility that some parts of this Standard may conflict with a patent right, application for a patent after opening to the public, utility model right or application for registration of utility model after opening to the public which have technical properties. The relevant Minister and the Japanese Industrial Standards Committee are not responsible for identifying the patent right, application for a patent after opening to the public, utility model right or application for registration of utility model after opening to the public which have the said technical properties.

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**Method for determination of
tetra-through octachlorodibenzo-p-dioxins,
tetra-through octachlorodibenzofurans
and dioxin-like polychlorinated biphenyls
in stationary source emissions**

1 Scope This Japanese Industrial Standard specifies the method of analysis for tetra-through octachlorodibenzo-para-dioxins, tetra-through octachlorodibenzofurans and dioxin-like PCBs (hereafter referred to as "dioxins") in exhaust gas which are generated by combustion, chemical reactions and discharged to a flue, stack or duct (hereafter referred to as "duct") in stationary source emissions using a gas chromatograph coupled to a mass spectrometer (hereafter referred to as "GC/MS"). The GC/MS employed in this Standard shall be the double-focusing mass spectrometer (MS) where the capillary column of a gas chromatograph (GC) is used and its resolution is 10 000 or more.

The minimum limits of detection of the GC/MS in this Standard, though it is liable to vary in accordance with an apparatus or analytical conditions, are 0.1 pg for tetrachlorinated compounds and pentachlorinated compounds, 0.2 pg for hexachlorinated compounds and heptachlorinated compounds, 0.5 pg for octachlorinated compounds and 0.2 pg or under for dioxin-like PCBs.

2 Normative references The following standards contain provisions which, through reference in this Standard, constitute provisions of this Standard. The most recent editions of the standards (including amendments) indicated below shall be applied.

- JIS K 0095 *Methods for sampling of flue gas*
- JIS K 0114 *General rules for gas chromatographic analysis*
- JIS K 0123 *General rules for analytical methods in gas chromatography mass spectrometry*
- JIS K 0211 *Technical terms for analytical chemistry (General part)*
- JIS K 0215 *Technical terms for analytical chemistry (analytical instrument part)*
- JIS K 0301 *Methods for determination of oxygen in flue gas*
- JIS K 0557 *Water used for industrial water and wastewater analysis*
- JIS K 0901 *Form, size and performance testing methods of filtration media for collecting airborne particulate matters*
- JIS K 1107 *High purity nitrogen*
- JIS K 8040 *Acetone for pesticide residue and polychlorinated biphenyl analysis*
- JIS K 8105 *Ethylene glycol*
- JIS K 8117 *Dichloromethane for pesticide residue and polychlorinated biphenyl analysis*
- JIS K 8180 *Hydrochloric acid*
- JIS K 8550 *Silver nitrate*

- JIS K 8574 *Potassium hydroxide*
JIS K 8680 *Toluene*
JIS K 8825 *Hexane for pesticide residue and polychlorinated biphenyl analysis*
JIS K 8891 *Methanol*
JIS K 8951 *Sulfuric acid*
JIS K 8987 *Sodium sulfate*
JIS K 9702 *Dimethyl sulfoxide*
JIS K 9703 *2,2,4-Trimethylpentane*
JIS R 3503 *Glass apparatus for chemical analysis*
JIS R 3505 *Volumetric glassware*
JIS Z 8401 *Guide to the rounding of numbers*
JIS Z 8808 *Methods of measuring dust concentration in flue gas*

3 Definitions For the purposes of this Standard, the definitions given in **JIS K 0095**, **JIS K 0114**, **JIS K 0123**, **JIS K 0211** and **JIS K 0215**, and the following definitions apply.

- a) **dioxins** a generic term of tetra-through octachlorodibenzo-para-dioxins, tetra-through octachlorodibenzofurans and dioxin-like PCBs
- b) **isomer** each compound whose number of chlorines substituent is the same, but their positions are different
- c) **homologue** a group of compounds whose number of chlorines substituent is the same, but their positions are different
- d) **PCDDs** Polychlorodibenzo-para-dioxins
- e) **PCDFs** Polychlorodibenzofurans
- f) **TeCDDs** Tetrachlorodibenzo-para-dioxins
- g) **PeCDDs** Pentachlorodibenzo-para-dioxins
- h) **HxCDDs** Hexachlorodibenzo-para-dioxins
- i) **HpCDDs** Heptachlorodibenzo-para-dioxins
- j) **OCDD** Octachlorodibenzo-para-dioxin
- k) **TeCDFs** Tetrachlorodibenzofurans
- l) **PeCDFs** Pentachlorodibenzofurans
- m) **HxCDFs** Hexachlorodibenzofurans
- n) **HpCDFs** Heptachlorodibenzofurans
- o) **OCDF** Octachlorodibenzofuran
- p) **isomers substituted by chlorine at 2,3,7,8-positions** Total of 17 types of isomers substituted by chlorine at 2,3,7,8-positions as shown below. They are composed of 7 types of tetra-through octachlorodibenzo-para-dioxins, and 10 types of tetra-through octachlorodibenzofurans.

- 1) **Tetra-to octachlorodibenzo-para-dioxins (PCDDs)**
 - 2,3,7,8-Tetrachlorodibenzo-para-dioxin (2,3,7,8-TeCDD)
 - 1,2,3,7,8-Pentachlorodibenzo-para-dioxin (1,2,3,7,8-PeCDD)
 - 1,2,3,4,7,8-Hexachlorodibenzo-para-dioxin (1,2,3,4,7,8-HxCDD)
 - 1,2,3,6,7,8-Hexachlorodibenzo-para-dioxin (1,2,3,6,7,8-HxCDD)
 - 1,2,3,7,8,9-Hexachlorodibenzo-para-dioxin (1,2,3,7,8,9-HxCDD)
 - 1,2,3,4,6,7,8-Heptachlorodibenzo-para-dioxin (1,2,3,4,6,7,8-HpCDD)
 - Octachlorodibenzo-para-dioxin (OCDD)
- 2) **Tetra-to octachlorodibenzofurans (PCDFs)**
 - 2,3,7,8-Tetrachlorodibenzofuran (2,3,7,8-TeCDF)
 - 1,2,3,7,8-Pentachlorodibenzofuran (1,2,3,7,8-PeCDF)
 - 2,3,4,7,8-Pentachlorodibenzofuran (2,3,4,7,8-PeCDF)
 - 1,2,3,4,7,8-Hexachlorodibenzofuran (1,2,3,4,7,8-HxCDF)
 - 1,2,3,6,7,8-Hexachlorodibenzofuran (1,2,3,6,7,8-HxCDF)
 - 1,2,3,7,8,9-Hexachlorodibenzofuran (1,2,3,7,8,9-HxCDF)
 - 2,3,4,6,7,8-Hexachlorodibenzofuran (2,3,4,6,7,8-HxCDF)
 - 1,2,3,4,6,7,8-Heptachlorodibenzofuran (1,2,3,4,6,7,8-HpCDF)
 - 1,2,3,4,7,8,9-Heptachlorodibenzofuran (1,2,3,4,7,8,9-HpCDF)
 - Octachlorodibenzofuran (OCDF)
- q) **PCBs** Polychlorobiphenyls
- r) **TeCBs** Tetrachlorobiphenyls
- s) **PeCBs** Pentachlorobiphenyls
- t) **HxCBs** Hexachlorobiphenyls
- u) **HpCBs** Heptachlorobiphenyls
- v) **dioxin-like PCBs (DL-PCBs)** polychlorobiphenyls (PCBs) which include compounds where ortho positions (2,2',6 and 6') are not substituted by chlorine (non-ortho compounds) and compounds having one substituted chlorine at the ortho position (mono-ortho compounds) That is also called coplanar PCB.
 - 1) **non-ortho compounds**
 - 3,4,4',5-Tetrachlorobiphenyl [3,4,4',5-TeCB (IUPAC* No. 81)]
 - Note * Abbreviation of International Union of Pure and Applied Chemistry.
 - 3,3',4,4'-Tetrachlorobiphenyl [3,3',4,4'-TeCB (IUPAC No. 77)]
 - 3,3',4,4',5-Pentachlorobiphenyl [3,3',4,4',5-PeCB (IUPAC No. 126)]
 - 3,3',4,4',5,5'-Hexachlorobiphenyl [3,3',4,4',5,5'-HxCB (IUPAC No. 169)]
 - 2) **mono-ortho compounds**
 - 2,3,3',4,4'-Pentachlorobiphenyl [2,3,3',4,4'-PeCB (IUPAC NO. 105)]
 - 2,3,4,4',5-Pentachlorobiphenyl [2,3,4,4',5-PeCB (IUPAC No. 114)]

- 2,3',4,4',5-Pentachlorobiphenyl [2,3',4,4',5-PeCB (IUPAC NO. 118)]
- 2',3,4,4',5-Pentachlorobiphenyl [2',3,4,4',5-PeCB (IUPAC No. 123)]
- 2,3,3',4,4',5-Hexachlorobiphenyl [2,3,3',4,4',5-HxCB (IUPAC No. 156)]
- 2,3,3',4,4',5'-Hexachlorobiphenyl [2,3,3',4,4',5'-HxCB (IUPAC No. 157)]
- 2,3',4,4',5,5'-Hexachlorobiphenyl [2,3',4,4',5,5'-HxCB (IUPAC No. 167)]
- 2,3,3',4,4',5,5'-Heptachlorobiphenyl [2,3,3',4,4',5,5'-HpCB (IUPAC No. 189)]

- w) **the minimum limit of detection for the apparatus** the minimum value that can be detected with the GC/MS used for the measurement
- x) **the minimum limit of detection for the analytical method** the minimum value that can be detected in the procedure of the operation from pretreatment of the sample to the measurement with GC/MS
- y) **the minimum limit of detection for the sample gas** the minimum concentration that can be detected for the sample gas
- z) **the minimum limit of the determination for the apparatus** the minimum value that can be determined by the GC/MS used for the measurement
- aa) **the minimum limit of the determination for the analytical method** The minimum value which can be determined in the procedure of the operation from pretreatment of the sample to the measurement with GC/MS. In general, approximately 3-fold errors may be estimated for the value in vicinity of the minimum limit of the detection, to compare with the values obtained near the minimum limit of the determination.
- ab) **minimum limit of determination for the sample gas** the minimum concentration that can be determined in the sampling gas
- ac) **TEF** 2,3,7,8-TeCDD toxic equivalency factor
- ad) **TEQ** 2,3,7,8-TeCDD toxic equivalent quantity
- ae) **RRF** relative response factor

4 Outline of the analytical method The dioxins in exhaust gas shall be collected by "collection filter" with using a cylindrical filter and such like, "absorption collection" with using a linked absorption bottle (impinger) and "adsorption collection" by an adsorption column. After extracting them from collection part, they shall be cleaned up, identified and determined using a GC/MS. The flow diagram of the analysis is shown in figure 1.

Remarks : Because dioxins are highly toxic, inhalation, accidental ingestion or direct dermal contact with them shall be avoided as much as possible. The pretreatment room and laboratory room shall be well-ventilated, and the management of waste water and wastes shall also be sufficiently carried out. Inhalation or accidental ingestion of other reagents, solvents, etc. may be detrimental to the health of technicians and for this reason, the utmost care shall be taken when handling them and the laboratory shall be sufficiently ventilated.

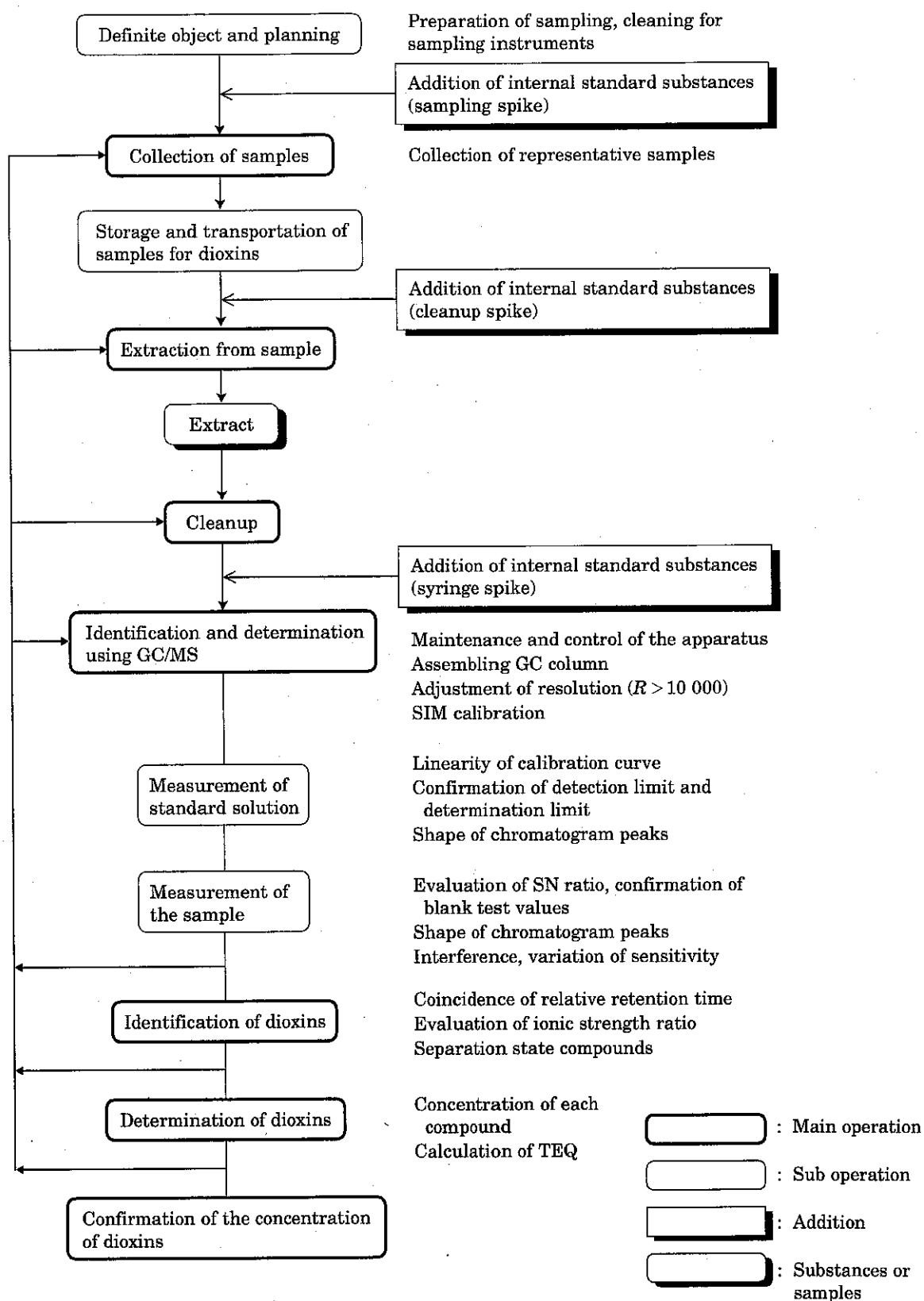


Figure 1 Flow diagram for the determination of dioxins in exhaust gas

5 Gas sampling

General matters of the gas sampling shall be in accordance with JIS K 0095.

5.1 Outline of gas sampling

Outline for the procedures of the gas sampling is shown in figure 2.

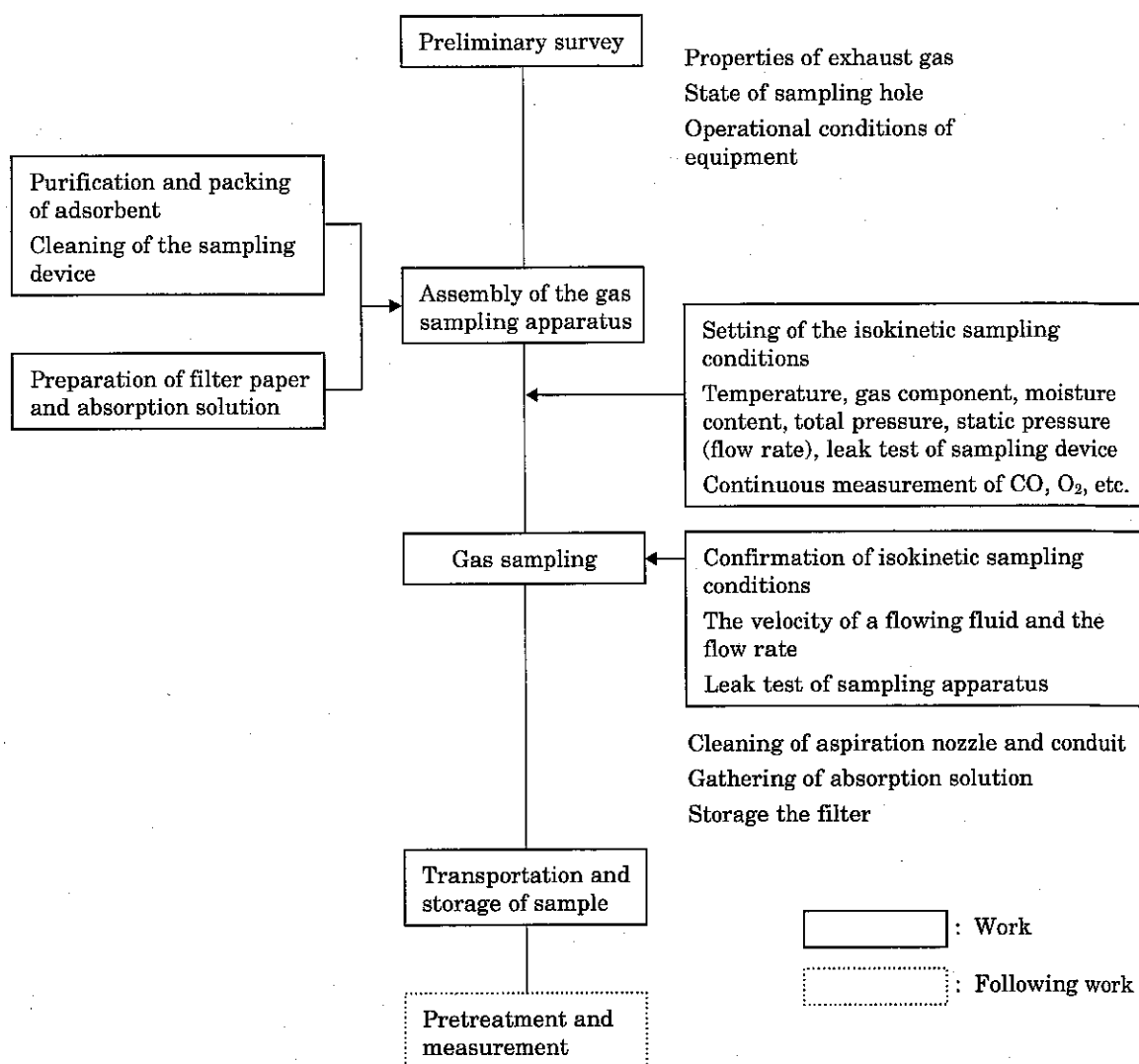


Figure 2 Sampling procedures for collecting gas samples

5.2 Gas sampling apparatus

Gas sampling apparatus shall be composed of a sampling pipe part, a collecting part, a suction pump and a flow rate meter. The schematic diagram of a sampling apparatus and a sampling pipe part are shown in figure 3 and figure 4 respectively.

Gas sampling apparatus shall provide the following conditions.

- Gas sampling by the isokinetic sampling shall be possible in the range of -5 % to +10 % of relative error for the exhaust gas flow rate at a measuring point.
- Sufficient high collection efficiency shall be obtained for gas sampling of dioxins.

- There is no possibility of secondary formation or decomposition of dioxins.
- There is no loss of dioxins in the operation till extracting process of them after sampling.
- There is no contamination due to dust and no mixing of the atmosphere at the site into the equipment while sampling.

An example of sampling apparatus is as shown in annex 1. The apparatus shown in annex 1 is that which has already the actual results developed for dioxins or that which is sufficiently confirmed to satisfy the above-mentioned conditions. Even in the case of the apparatus other than that as shown herein, when the following conditions are confirmed about at least the three kinds of exhaust gas of different composition, that method may be used⁽¹⁾ within the range of the composition of exhaust gas used for confirmation.

- The flow rate of suction calculated from the flow rate of exhaust gas at measuring point is within the range of the adjustment of suction flow rate of apparatus.
- No leakage in the apparatus.
- One more stage of the part capable of collecting dioxins at the back of sampling apparatus, for example, the adsorption collecting part of JIS type I of sampling apparatus in annex 1 is added and the sample is taken, and dioxins are not detected from the collecting part added.
- The same sample gas is taken in parallel with JIS type I of sampling apparatus in annex 1 simultaneously, and the concentration of the dioxins conforms within $\pm 30\%$.

Note (1) The composition of exhaust gas used for confirmation means temperature, moisture, flow rate, oxygen concentration, dust concentration and dioxins concentration and when even any one of them differs by not less than 30 %, the composition is considered to be different. And, the phrase of within the range of composition means that $\pm 30\%$ of each value is made the limit.

- a) **Sampling pipe part** Sampling pipe shall be used that made of the borosilicate glass or vitreous silica depending on temperature of exhaust gas. When temperature of the dust collecting part in sampling apparatus cannot maintain to 120 °C or lower, a cooling probe with water cooled tube shall be employed. The following requirements shall be satisfied for the sampling pipe.
 - 1) Gas flow inside and outside of the pipe shall not be turbulent. Inside diameter of a nozzle shall be at least 4 mm and measured with a 0.1 mm accuracy.
 - 2) The tip of the nozzle shall be finished to a sharpness angle of 30° or less, or a smooth hemisphere. Inside and outside of the nozzle shall be smooth.
 - 3) The sampling pipe shall be smooth for the inside and outside from the nozzle to the dust collecting part with no acute profile change or bend.
- b) **Collecting part** The collecting part is the part collecting in good efficiency the dioxins in gas condition or adsorbed on dust. There are the collection by filter, the collection by liquid, the collection by adsorbent, etc. and the dioxins are collected by their combination.

- c) **Connecting part** For the connecting pipe from the sampling pipe part up to the part cooled at the dew point or lower of sampling gas, glass pipe shall be used. For the connecting pipe and connection of other part than that, the pipe of glass or fluorocarbon resin shall be used and grease shall not be used at connection part.
- d) **Suction pump** The pump shall be capable of aspirating a flow rate of 10 L/min to 40 L/min and have a flow rate regulating function. It shall be capable of more than 24 h continuous operation.
- e) **Flow rate meter** A wet type or dry type gas meter shall be employed as a designated flow meter. This shall be capable of measuring between 10 L/min to 40 L/min with 0.1 L/min increment. The scale of the designated flow meter shall be calibrated at regular intervals by requesting to the manufacturer and so on.

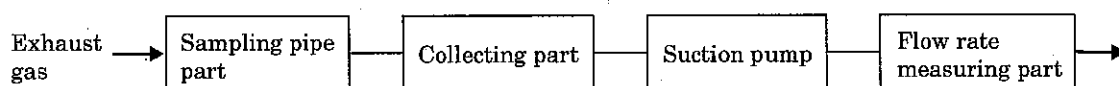


Figure 3 Schematic diagram of sampling apparatus for sample gas

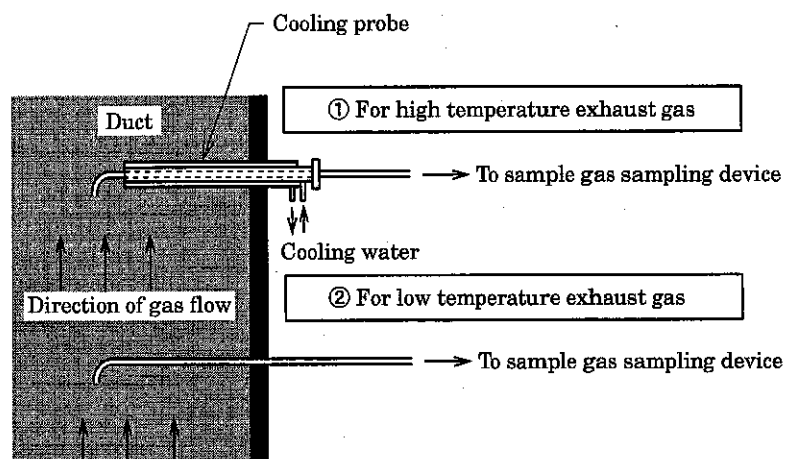


Figure 4 Schematic diagram of the sampling pipe part

5.3 Reagents The following reagents shall be used in the gas sampling. The reagents other than those required depending on sampling apparatus shall be in accordance with the specification of the apparatus. Blank test values and the like shall be used to confirm that these reagents do not affect the measurement.

- a) **Water** A4 (or A3) specified in **JIS K 0557**.
- b) **Acetone** As specified in **JIS K 8040**, or equivalent grade.
- c) **Toluene** As specified in **JIS K 8680**, or equivalent grade.
- d) **Hexane** As specified in **JIS K 8825**, or equivalent grade.
- e) **Methanol** As specified in **JIS K 8891**, or equivalent grade.
- f) **Dichloromethane** As specified in **JIS K 8117**, or equivalent grade.

- g) **Hexane washing water** Water at a) rinsed thoroughly by hexane at d).
- h) **Internal standard substances for sampling spike** Internal standard substance added to check the results of the procedures from sampling through extraction. A solution shall be used that contains at least one appropriate type of PCDDs, PCDFs and PCBs labelled with ^{13}C or ^{37}Cl and diluted to the appropriate concentration using toluene or nonane.

In the internal standard substances, except for above spike, there are 2 types of spikes. One of the spikes is used as the cleanup spike, which is added to check the results of all pretreatment operations from the extraction to cleanup and employing as the standard substances for determination of dioxins. The other spike is used as the syringe spike, which is added to check the injection of a sample solution into a GC/MS. The internal standard substances for cleanup spike shall be used, at least for each number of chlorine, the minimum one type of isomers substituted by chlorine at 2,3,7,8-positions, the internal standard substances for syringe spike shall be used the internal standard substances except for the sampling spike and the cleanup spike used. An example of the internal standard substances for dioxins is shown in table 1 (see annex 2).

Table 1 Examples of the internal standard substances for dioxins

		Compound
PCDDs	Tetrachlorinated compounds	$^{13}\text{C}_{12}$ -1,2,3,4-TeCDD
		$^{13}\text{C}_{12}$ -2,3,7,8-TeCDD
		$^{37}\text{Cl}_4$ -2,3,7,8-TeCDD
	Pentachlorinated compounds	$^{13}\text{C}_{12}$ -1,2,3,7,8-PeCDD
	Hexachlorinated compounds	$^{13}\text{C}_{12}$ -1,2,3,4,7,8-HxCDD
		$^{13}\text{C}_{12}$ -1,2,3,6,7,8-HxCDD
		$^{13}\text{C}_{12}$ -1,2,3,7,8,9-HxCDD
	Heptachlorinated compounds	$^{13}\text{C}_{12}$ -1,2,3,4,6,7,8-HpCDD
	Octachlorinated compounds	$^{13}\text{C}_{12}$ -1,2,3,4,6,7,8,9-OCDD
PCDFs	Tetrachlorinated compounds	$^{13}\text{C}_{12}$ -2,3,7,8-TeCDF
		$^{13}\text{C}_{12}$ -1,2,7,8-TeCDF
	Pentachlorinated compounds	$^{13}\text{C}_{12}$ -1,2,3,7,8-PeCDF
		$^{13}\text{C}_{12}$ -2,3,4,7,8-PeCDF
	Hexachlorinated compounds	$^{13}\text{C}_{12}$ -1,2,3,4,7,8-HxCDF
		$^{13}\text{C}_{12}$ -1,2,3,6,7,8-HxCDF
		$^{13}\text{C}_{12}$ -1,2,3,7,8,9-HxCDF
		$^{13}\text{C}_{12}$ -2,3,4,6,7,8-HxCDF
	Heptachlorinated compounds	$^{13}\text{C}_{12}$ -1,2,3,4,6,7,8-HpCDF
		$^{13}\text{C}_{12}$ -1,2,3,4,7,8,9-HpCDF
	Octachlorinated compounds	$^{13}\text{C}_{12}$ -1,2,3,4,6,7,8,9-OCDF
PCBs*	Tetrachlorinated compounds	$^{13}\text{C}_{12}$ -2,3',4',5-TeCB (#70)
		$^{13}\text{C}_{12}$ -3,3',4,4'-TeCB (#77)**
		$^{13}\text{C}_{12}$ -3,4,4',5-TeCB (#81)**
	Pentachlorinated compounds	$^{13}\text{C}_{12}$ -2,3,3',4,4'-PeCB (#105)**
		$^{13}\text{C}_{12}$ -2,3,4,4',5-PeCB (#114)**
		$^{13}\text{C}_{12}$ -2,3',4,4',5-PeCB (#118)**
		$^{13}\text{C}_{12}$ -2',3,4,4',5-PeCB (#123)**
		$^{13}\text{C}_{12}$ -3,3',4,4',5-PeCB (#126)**
	Hexachlorinated compounds	$^{13}\text{C}_{12}$ -2,3,3',4,4',5-HxCB (#156)**
		$^{13}\text{C}_{12}$ -2,3,3',4,4',5'-HxCB (#157)**
		$^{13}\text{C}_{12}$ -2,3,4,4',5,5'-HxCB (#167)**
		$^{13}\text{C}_{12}$ -3,3',4,4',5,5'-HxCB (#169)**
	Heptachlorinated compounds	$^{13}\text{C}_{12}$ -2,2',3,3',4,4',5-HpCB (#170)
		$^{13}\text{C}_{12}$ -2,2',3,4,4',5,5'-HpCB (#180)
		$^{13}\text{C}_{12}$ -2,3,3',4,4',5,5'-HpCB (#189)**

Notes * Numerical values in parentheses show the number of IUPAC.

** That indicates DL-PCB.

5.4 Preparations for sampling of sample gas

5.4.1 Preliminary survey The properties of exhaust gas which is discharged from the incineration plant measured are differ depending on factors such as the scale of operation and the exhaust gas treatment method. Many measurement sites may be dangerous to operate in. For these reasons, it is necessary to survey previously at the measuring sites in order to check the properties of exhaust gas and the security for work. The following items shall be included in this preliminary survey.

The sampling position of exhaust gas shall be the position where exhaust gas with typical properties is collected and, among the measuring positions of flow rate specified in clause 4 of **JIS Z 8808**, the point (the feasible spot of the isokinetic sampling) where is as close as possible to the point that the average velocity is obtained.

- a) **Properties of the exhaust gas** Temperature, flow rate, composition, pressure, moisture content and so on.
- b) **Sampling position** Height above ground, state of the sampling hole, positions of the blower and air-exhaust ventilator and so on.
- c) **Duct** Shape, size (dimensions), etc. of duct.
- d) **Security of working** Extents of a sampling stage, conditions of a ladder and so on.
- e) **Power source, water supply** Existence of power source and water supply, or not.
- f) **Others**

5.4.2 Preparation for the equipment Based on the results of the preliminary survey, the required apparatus for gas sampling shall be selected and prepared to meet the conditions of the sampling site and carried out the following preparations.

- a) Apparatus required for collecting dusts.
- b) Absorption bottles, adsorbent column, absorption liquids and other devices required for collecting dioxins in the exhaust gas.
- c) Ice or dry ice for cooling.
- d) Reagents required to wash the collecting unit after gas sampling [methanol⁽²⁾, dichloromethane⁽³⁾].

Notes (2) Acetone may also be used.

(3) Toluene may also be used.

- e) **Others**

5.4.3 Addition of the internal standard substance (sampling spike) The addition of the internal standard substance before gas sampling refers to as a sampling spike and the internal standard substance for a sampling spike is added in dioxins collecting part⁽⁴⁾.

Different isomers are used as the sampling spike, the cleanup spike and the syringe spike respectively. More than one appropriate type of sampling spike is used. Because some internal standard substances may affect the analysis depending on the conditions of a mass spectrometer, it is needed to examine and confirm sufficiently when such internal standard substances used [refer to 6.4.1 note (8), refer to annex 2]. Recovery ratio of the internal standard substances of sampling spike shall be between 70 % and 130 %.

Note (4) Depending on the concentrations and the measuring conditions of exhaust gas, amounts of internal standard substances to be added shall be determined so that the concentrations are put in the range of the calibration curve of the measurement with GC/MS. Generally, approximately 1 ng to 20 ng shall be added.

5.5 Sampling amount of sample gas The sampling amount of sample gas shall be determined according to the following procedures. For the sampling time, the representative sample shall be made to take as considering sufficiently the generating condition of sample gas corresponding to the purpose.

- a) Determine the minimum concentration to be evaluated from the purpose of measurement.
- b) Unless especially specified, set up the minimum limit of detection in sample gas at not more than 1/30 of the concentration determined in a).
- c) Calculate the minimum amount of sample gas required for measurement according to formula (1).

$$V = \frac{Q_{DL}}{1\,000} \times \frac{y}{x} \times \frac{V_E}{V'_E} \times \frac{1}{C_{DL}} \dots\dots\dots (1)$$

where, V : minimum amount of sample gas required for measurement (0 °C, 101.32 kPa) (m³)
 Q_{DL} : minimum limit of detection of measurement method (pg)
 y : liquid amount of sample for measurement (μl)
 x : injection amount to GC/MS (μl)
 V_E : amount of extracted liquid (ml)
 V'_E : aliquot amount of extracted liquid (ml)
 C_{DL} : minimum limit of detection of sample gas to be required (0 °C, 101.32 kPa) (ng/m³)

- d) Take the amount of not less than the minimum amount of sample gas calculated as the sampling amount of sample gas.

5.6 Sampling procedure Procedure for gas sampling shall be as follows.

- a) Temperature of exhaust gas, flow rate, pressure, moisture contents, etc. shall be measured according to **JIS Z 8808** and calculate the flow rate of exhaust gas at the measuring point⁽⁵⁾.

Note (5) When the continuous measurements of carbon monoxide, oxygen, etc. are carried out simultaneously, they shall be continued from at least 1 h before the start of the dioxin sampling till the end of that unless otherwise stated, and the operation conditions shall be ascertained at the same time.

- b) Calculate the flow rate of suction by considering the sampling amount of sample gas, sampling time, and determine the inside diameter of nozzle so as to be isokinetic sampling.
- c) The leakage test, with assembled sampling apparatus, shall be carried out. Leakage shall be confirmed by checking whether the needle of the gas meter does not move when the nozzle in the sampling pipe is closed and then the suction pump is operated. The measurement result shall be recorded.

- d) Insert a nozzle of the sampling pipe of which the direction is the opposite against the gas flow through the sampling hole to the sampling point. Read the indicated value on the gas meter. Face the direction of the nozzle accurately toward the flow of the exhaust gas at the same time of operating the suction pump and suck the exhaust gas by the isokinetic sampling⁽⁶⁾. At the time, the following points shall be taken into consideration.

Note ⁽⁶⁾ When the concentration of dust in exhaust gas is 1 mg/m³ (0 °C, 101.32 kPa) or under, the constant suction at about average flow rate, without carrying out isokinetic sampling is allowed, and when the isokinetic sampling is not applicable, it may not be applied as well.

- 1) The flow rate aspirated from the nozzle of the sampling pipe shall be within a relative error range of -5 % to +10 % against that of exhaust gas at the measuring point. It is desirable to measure the flow rate of exhaust gas every 60 min and adjust the isokinetic sampling rate. When isokinetic sampling is difficult, and the suction flow rate is reduced, pause suction and exchange the filter of the collecting part and the like.
 - 2) The leakage test for the sampling apparatus shall be carried out at least once during gas sampling. In this case, no leakage shall be confirmed when no difference is ensured between the concentration of oxygen obtained at a sample point and that obtained at the outlet of the pump. The test result shall be recorded. The measurement of oxygen concentration shall be in accordance with JIS K 0301. In the case of the detaching of line because of the exchange of filtering material in filter-collecting part, draining of drain, etc., carry out the test surely after reversion.
 - 3) If exhaust gas temperature is high and there is a possibility that the temperature at the dust collecting part exceeds 120 °C, use air cooling, etc. to cool the sampling pipe to below 120 °C. Furthermore, in some cases, use cooling probe as shown in figure 4 to ensure that the temperature at the dust collecting part does not exceed 120 °C.
 - 4) Immerse the collecting part which needs to be cooled in water or dry ice bath during exhaust gas sampling. The dioxins collecting part shall be kept away from direct light.
 - 5) For others, follow the respective precautions every collecting apparatus.
- e) Record the gas meter temperature and pressure.
- f) Once the volume of the sample gas has been sucked, return the nozzle of the sampling pipe to face the opposite direction, stop the suction pump, read the indication of needle in the gas meter gauge. Take out the sampling pipe.

If the pressure inside the duct is negative, take out the sampling pipe promptly from the duct while operating the suction pump, and then stop the pump.

5.7 Recovery and storage of the sample When the gas sampling has been completed, the gas sampling apparatus shall be decomposed only parts of the minimum requirement. Avoiding to mix with air and shielding from the direct light, the apparatus shall be transported into the laboratory. Remove each part of the gas sampling apparatus with care. Wash the sampling pipe and conduit pipe thoroughly with

methanol⁽²⁾, or dichloromethane⁽³⁾. Transfer the washing solution to a brown bottle with the sampling solution and store it. Put the filters, adsorbent and the like into a container and store them in a dark storage. For the samples stored, the following operations after the pretreatment shall be immediately carried out.

When transporting the samples, special care shall be taken not to break the containers and not to lose any of the solvents or sample components through volatilization.

5.8 Calculation of gas sampling volume Volume of dry gas sucked under the normal state shall be calculated according to formula (2).

$$V_{SD} = V_m \times \frac{273.15}{273.15 + t} \times \frac{P_a + P_m - P_v}{101.32} \times 10^{-3} \dots\dots\dots (2)$$

where, V_{SD} : gas sampling volume under the normal state (0 °C, 101.32 kPa) (m³)

V_m : reading of the gas meter (L)

t : temperature of sampling gas on the gas meter (°C)

P_a : atmospheric pressure (kPa)

P_m : gauge pressure of sampling gas on the gas meter (kPa)

P_v : vapour pressure of water at t °C (kPa)

If a dry gas meter is employed, and gas has been dried before the meter previously, the term P_v in the formula can be omitted.

5.9 Report of gas sampling When sampling of gas has been carried out, the following information shall be arranged and recorded. If needed, photographs on the site shall be taken.

- a) **Date and time of gas sampling**
- b) **Conditions of gas sampling location** Type of sources, working conditions, sampling position, conditions of vicinity, schematics, etc.
- c) **Conditions and situation of the sampling target** Temperature, moisture contents, static pressure, flow rate, wet and dry flow rate, colouring of sampling system and so on.
- d) **Gas sampling conditions** Configuration of gas sampling equipment, the results of leakage test, isokinetic sampling flow rate, sampling times, sampling volumes and collecting dust volume if needed.

6 Pretreatment of the sample

6.1 Outline for the pretreatment of the sample After the addition of the internal standard substance (cleanup spike) to the sample, the extractions shall be carried out for filter, resin and absorption solution separately and combine them together. Take an aliquot of the solution as required, treat the fluid with sulfuric acid—silica gel column chromatograph or multilayer silica gel column chromatograph⁽⁷⁾. Then, any one of the operation of alumina column chromatograph, the operation of

high performance liquid chromatograph, the operation of activated carbon column chromatograph or these combination of refining operation shall be carried out. When the oil component such as mineral oil is much in the sample, the operation of dimethylsulfoxide partition treatment may be added to the refining operation as occasion demands. After the refining operation the samples are analyzed using GC/MS. The summary of refining operation is shown in table 2. Figure 5 shows one of the examples of analytical flow chart of a sample.

Note (7) The operation of sulfuric acid treatment—multilayer silica gel chromatograph is allowed.

Table 2 Summary of refining operation

Name of operation	Main effect
Sulfuric acid treatment—operation of silica gel column chromatograph	Decomposition-removal of the greater part of matrix Removal of coloured substance, polynuclear aromatic hydrocarbon, high polar substance
Operation of multilayer silica gel column chromatograph	Removal of phenols, acidic substance, lipid, protein, sulfur compound, aliphatic hydrocarbons, high polar substance, coloured substance, polynuclear aromatic hydrocarbon
Operation of alumina column chromatograph	Removal of low polar substance, organic chlorine compound
Operation of high performance liquid chromatograph	Separation-refining of PCDDs, PCDFs, dioxin-like PCB
Operation of activated carbon column chromatograph	Separation-refining of PCDDs, PCDFs, dioxin-like PCB
Operation of dimethylsulfoxide (DMSO) partition treatment	Removal of low polar substance such as aliphatic hydrocarbon

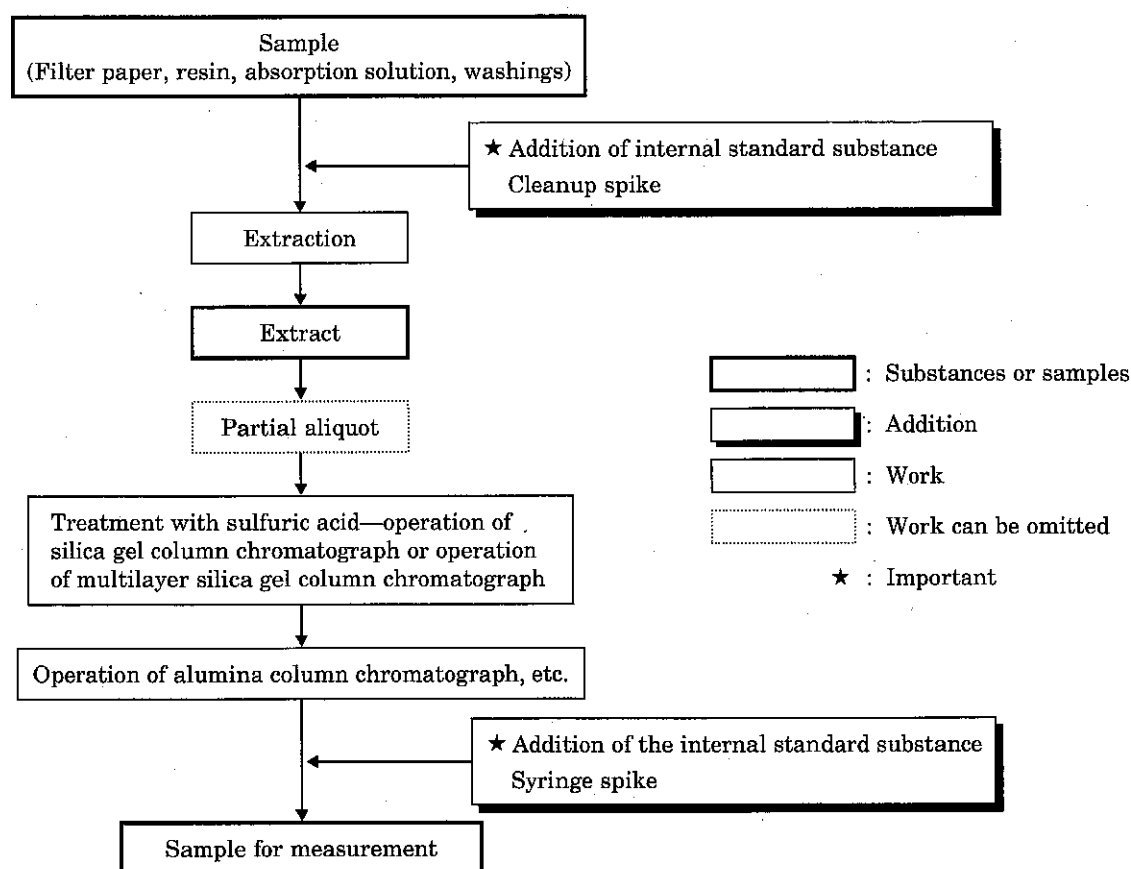


Figure 5 An example of a flow diagram from the pretreatment of a sample to its measurement

Moreover, even the operation other than the refining operation mentioned herein may be used when it is confirmed to satisfy the following conditions. For this confirmation, the data of 25 points or more in total shall be required about the sample medium to be applied so as to repeat five times or more by using five or more samples at different sampling point.

- a) The yield of dioxins to be object shall be not less than 90 %.
- b) The sample solution obtained by means of the refining operation specified in this Standard and the sample solution obtained by means of the new operation method to be adopted are measured by using the low revolution of GC/MS such as quadruple pole type according to the total ion detection method in the range of 50 to 450 in measurement mass number under the condition of gas chromatograph in a) of 7.3.1, and it shall be confirmed that the difference is not found in refining effect as compared with the respective chromatographs obtained or the equivalent or over effect to the refining operation of this Standard can be obtained.
- c) The measurement in 7.3.3 is carried out for the sample solution obtained by means of the new operation method to be adopted and it shall be confirmed that variation is not found in the monitor channel of the reference material for correction of mass in the neighbourhood where the peak due to objective component for analysis appears.

6.2 Reagents The following reagents shall be used in the pretreatment of the sample. Blank test values, etc. shall be used to confirm that these reagents do not affect the measurement.

- a) **Water** A4 (or A3) specified in **JIS K 0557**.
- b) **Methanol** As specified in **JIS K 8891**, or equivalent grade.
- c) **Acetone** As specified in **JIS K 8040**, or equivalent grade.
- d) **Toluene** As specified in **JIS K 8680**, or equivalent grade.
- e) **Dichloromethane** As specified in **JIS K 8117**, or equivalent grade.
- f) **Hexane** As specified in **JIS K 8825**, or equivalent grade.
- g) **Dimethylsulfoxide** As specified in **JIS K 9702** or equivalent grade.
- h) **Decane** At a quality that does not affect the measurement.
- i) **Nonane** At a quality that does not affect the measurement.
- j) **2,2,4-trimethyl pentane** As specified in **JIS K 9703**, or equivalent grade.
- k) **Sodium sulfate** As specified in **JIS K 8987**.
- l) **Hydrochloric acid** Guaranteed grade as specified in **JIS K 8180**, or equivalent grade.
- m) **Sulfuric acid** As specified in **JIS K 8951**, or equivalent grade.
- n) **Potassium hydroxide** As specified in **JIS K 8574**, or equivalent grade.
- o) **Silver nitrate** As specified in **JIS K 8550**.
- p) **Hexane washing water** Water at a) rinsed thoroughly with hexane at f).
- q) **Internal standard substance for cleanup spike** A nonane or toluene solution shall be used that contains appropriate types and concentration of dioxins labelled with ^{13}C or ^{37}Cl for all carbon or chlorine atoms (refer to annex 2).
- r) **Internal standard substance for syringe spike** A nonane and others which the solvent is the same as the measuring sample shall be used that contains appropriate types and concentration of dioxins labeled with ^{13}C or ^{37}Cl for all carbon or chlorine atoms (refer to annex 2).
- s) **Silica gel** Place the silica gel (63 μm to 212 μm) for the column chromatograph in a beaker, wash with methanol and thoroughly evaporate the methanol. Transfer the silica gel into an evaporating dish or a beaker so that it forms a layer of 10 mm or less. After heating the gel for approximately 18 h at 130 °C, allow it to stand to cool for approximately 30 min in a desiccator. After preparation, transfer the gel into a reagent bottle which can be sealed and store it in a desiccator.
- t) **Silica gel impregnated with potassium hydroxide [2 % (mass fraction)]** After adding 40 ml of potassium hydroxide solution (50 g/L) preparing with potassium hydroxide of n) to 100 g of silica gel at s), dehydrate it under the reduced pressure at approximately 50 °C in a rotary evaporator. After removing

almost all the moisture, raise the temperature from 50 °C to 80 °C, continue the dehydration under the reduced pressure at this temperature for approximately 1 h to pulverize the silica gel. After preparation, place it in an air tight reagent bottle and store in a desiccator.

- u) **Silica gel impregnated with sulfuric acid [22 % (mass fraction)]** After adding 28.2 g of sulfuric acid at **m**) to 100 g of silica gel at **s**), shake the mixture sufficiently so that it forms a powder. After preparation, transfer the powdered mixture into a reagent bottle which can be sealed and store it in a desiccator.
- v) **Silica gel impregnated with sulfuric acid [44 % (mass fraction)]** After adding 78.6 g of sulfuric acid at **m**) to 100 g of silica gel at **s**), shake the mixture sufficiently so that it forms a powder. After preparation, transfer the powdered mixture into a reagent bottle which can be sealed and store it in a desiccator.
- w) **Silica gel impregnated with silver nitrate [10 % (mass fraction)]** After adding 28 ml of silver nitrate solution (400 g/L) prepared with silver nitrate at **o**) to 100 g of silica gel at **s**), remove the moisture thoroughly in a rotary evaporator. Silica gel impregnated with silver nitrate shall be, after preparation, transfer the mixture into a coloured bottle which can be sealed and store it in a desiccator.

In addition, light shall be shaded at the time of preparation and preservation as much as possible.

- x) **Alumina** If pre-activated alumina for use in column chromatograph (basic, activity grade I) is purchased from the supplier, then it can be used as it is. If activation is necessary, place 10 mm or less of alumina into a beaker and dry it at 130 °C for approximately 18 h, or put approximately 5 mm thickness of alumina into a Petri dish and dry it at 500 °C for approximately 8 h. Allow it to stand to cool for approximately 30 min in a desiccator and store in a reagent bottle which can be sealed. The alumina shall be used as soon as possible after activation.
- y) **Carbon column for high performance liquid chromatograph** Graphite carbon column for liquid chromatograph, or that having the equivalent separation performance to this.
Information 1 For example, Hypercarb (4.6 mm in inside diameter, 100 mm in length) manufactured by Hypersil Co., and so on are included.
- z) **Activated carbon column packing** Silica gel impregnated or dispersed with activated carbon or that having the equivalent separation performance to this.
Information 2 For example, activated carbon impregnated silica gel (WAKO Pure Chemical Industry Co., Ltd.), activated carbon dispersed silica gel (KANTO Chemical Co., Ltd.) and so on are included.
- aa) **Glass fiber filter** Pore size approximately 0.5 µm is preferable. The filter shall be used in a Büchner funnel.
Information 3 There are some goods called retained particle size instead of pore size on the market.
- ab) **Nitrogen** A high purity grade 2 of nitrogen specified in **JIS K 1107**.

6.3 Equipment and apparatus The equipment and apparatus used for pretreatment of the sample shall be washed sufficiently with methanol (or acetone) and toluene (or dichloromethane). Blank tests shall be conducted to confirm they will not affect the measurement.

- a) **Glassware** As specified in **JIS R 3503** and **JIS R 3505**. Those with cocks made of fluorine resin may be used.
- b) **Soxhlet extractor** As specified in **JIS R 3503** or equivalent. No sealant grease shall be used in the coupling section.
- c) **Concentrator** Kuderna-Danish (KD) concentrator or a rotary evaporator. No sealant grease shall be used in the coupling section.
- d) **Column-chromatograph tube** Glass tube for column-chromatograph with an inside diameter of 10 mm and a length of 300 mm (or an inside diameter of 15 mm and a length of 300 mm) without the adsorption and mixing of dioxins, the elution of interfering substance and so on, or equivalent quality of material to this.
- e) **Büchner funnel**
- f) **High performance liquid chromatograph**

6.4 Procedures for pretreatment

6.4.1 Addition of the internal standard substance (cleanup spike) A specific amount of the internal standard substance⁽⁸⁾ shall be added as a cleanup spike to the samples in the each collecting part. Amount to be added is generally, as total volume added in each unit, 0.1 ng to 2 ng for tetrachlorinated compounds to heptachlorinated compounds, 0.2 ng to 4 ng for octachlorinated compounds and 0.1 ng to 2 ng for DL-PCBs. When the concentration of dioxins in the sample is expected extremely high, and the total concentration is beyond the range of the determination only by the addition of the normal volume of the internal standard substance, an amount exceeding the determination range may also be added.

Note ⁽⁸⁾ For the internal standard substance for cleanup spike of PCDDs and PCDFs, at least one type of compound substituted by chlorine at 2,3,7,8-positions shall be added for each chlorine number. For PCBs, all types of non-ortho DL-PCBs, and one type of mono-ortho DL-PCBs or the other PCBs shall be added for each chlorinated compound. If possible, all of the compounds with toxicity equivalence factor should be added.

Although the internal standard substance to be added is different from the isomers for syringe spike, some of the internal standard substances may cause interference depending on the measuring conditions of GC/MS. Therefore, the internal standard substances have to be checked and confirmed thoroughly prior to use (see annex 2).

The recovery of the internal standard substance added as the cleanup spike shall be between 50 % and 120 % on the basis of the internal standard substance for syringe spike. If the recovery is out of this range, the pretreatment shall be repeated.

6.4.2 Extraction from each collecting part Carry out the extraction procedures for collecting part, combine with each extracted fluid and use it as the extracted solution. An example of the flow diagram to the preparation of extraction solution is shown in figure 6. The extraction procedures for each collecting part shall be as follows.

- a) **Extraction from solid** For the extraction from solid such as filtering material, adsorbent, the extraction⁽¹⁰⁾ shall be carried out by means of Soxhlet extraction or the equivalent extraction method⁽⁹⁾ to this. For the dust collected on filtering material, etc., after carrying out the following operation, the extraction operation shall be carried out.

Add hydrochloric acid (2 mol/L) so that 20 mmol or over of hydrochloric acid concentration for 1 g of dust shall be obtained. Allow it to stand to cool for approximately 1 h while confirming the foaming by occasional mixing. Then confirm that foaming does not occur by additional hydrochloric acid. Filter it through the glass fiber filter of approximately 0.5 μm pore-size in a Büchner funnel and wash thoroughly with hexane washing water. Wash it again by a little volume of methanol or acetone to remove the moisture and then dry it in air⁽¹¹⁾.

The filtrate shall be treated by liquid—liquid shaking extraction with dichloromethane and combined into the extracted fluid of the Soxhlet extractor.

Notes ⁽⁹⁾ The judgement of the equivalency to Soxhlet extraction method shall be carried out by judging whether the standard sample such as fly ash is measured and the measurement result conforms to the standard value or not, or whether the sample such as fly ash, etc. is measured in parallel to Soxhlet extraction and the measurement result conforms or not. For the judgement, nine data in total shall be used so as to repeat at least three times using three samples.

- ⁽¹⁰⁾ In the Soxhlet extraction, because the moisture content remaining in the sample may be lower extraction efficiency, it shall be removed using a suitable method before extraction. The application of a Soxhlet/Deanstag extractor is recommended (refer to EPA Method 1613 and others). And, care shall be taken to the decomposition by light because of requiring a long time operation.

- ⁽¹¹⁾ When air-drying, special care shall be taken to minimize dispersion or contamination of dioxins in the sample.

- b) **Extraction from liquid** The absorbed solution of aqueous solution and its washings are joined together and put in a separating funnel, dichloromethane is added at the rate of 100 ml per 1 L solution, and extracted by shaking at approximately 5 cm in amplitude, 100 or more r.p.m. for approximately 20 min. This extraction is carried out three times, the extracted solution is joined together and dehydrated by using sodium sulfate.

The organic solvent such as 2,2'-oxybisethanol (diethylene glycol) and its washings are transferred to a separating funnel, the same amount of hexane washings is added, dichloromethane is added at the rate of 100 ml per 1 L of solution, the liquid—liquid shaking extraction is similarly carried out three times, the extracted solution is joined together and dehydrated by using sodium sulfate.

6.4.3 Preparation of extract solution Combine the extracted fluids obtained by each collecting part, add the solvent and dilute to constant volume. This is used as the extract solution.

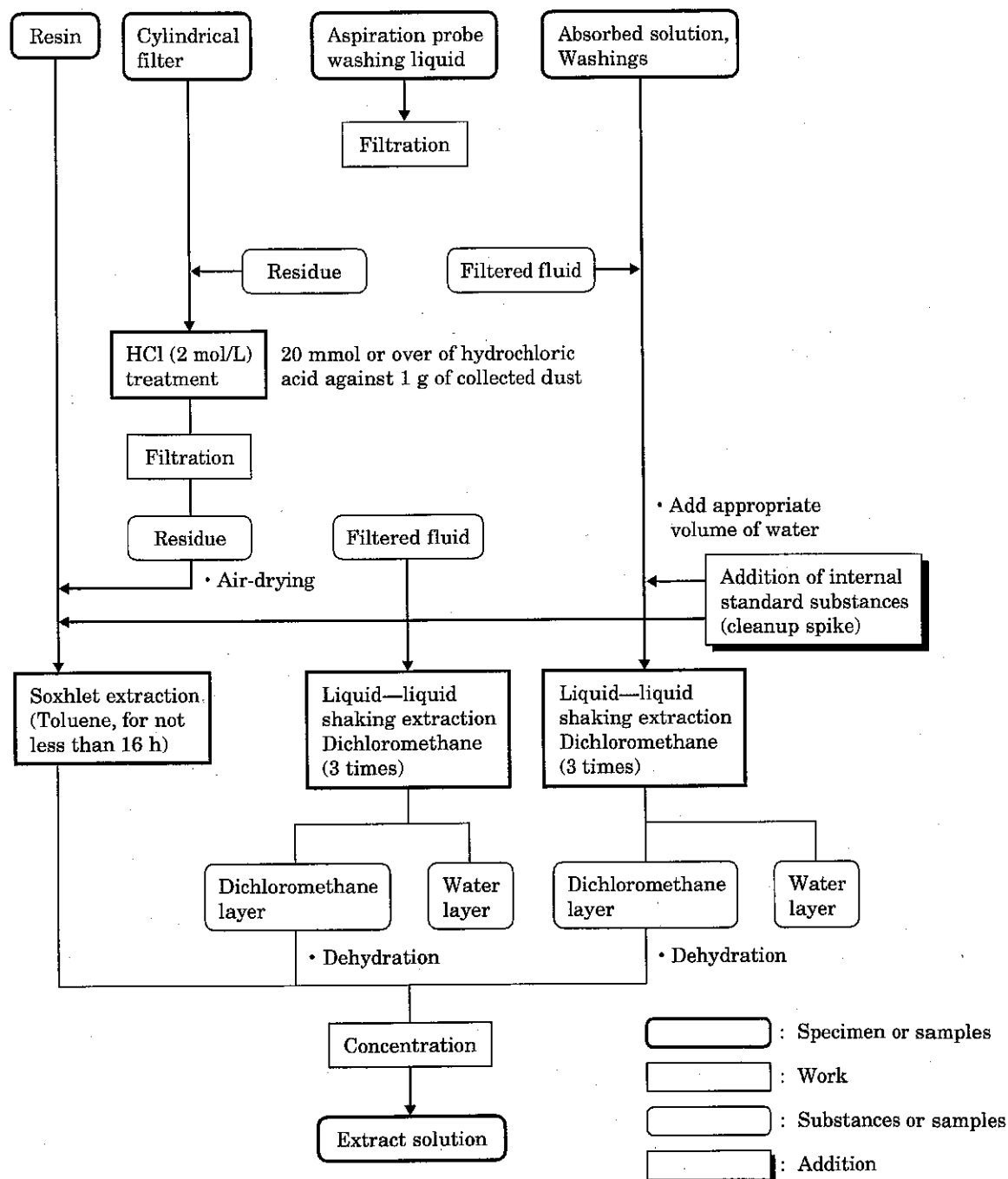


Figure 6 An example of a flow diagram to the preparation of extract solution

6.4.4 Operation of the sulfuric acid treatment—Silica gel column chromatograph or multilayer silica gel column chromatograph Take a suitable aliquot of the extract solution obtained in 6.4.3⁽¹²⁾, concentrate it to approximately 5 ml with

a concentrator. Then, remove toluene by nitrogen gas flow⁽¹³⁾ and make the volume to approximately 500 µl of concentrated solution. This concentrated solution shall be refined by operation of the sulfuric acid treatment—silica gel column chromatography or multilayer silica gel column chromatography.

Notes ⁽¹²⁾ Part of the extracted solution should be stored for re-analysis.

⁽¹³⁾ Special care shall be taken so that the target substances in analysis are not lost due to concentration with nitrogen gas flow, and that the solution is not scattered by adjusting nitrogen gas flow to the level that the surface of the solution is seen to show slight motion. The solution shall not be evaporated to dryness. Over-drying of the solution may cause loss of the target components.

a) **Operation of the sulfuric acid treatment—Silica gel column chromatograph** Operation of the sulfuric acid treatment—Silica gel column chromatograph shall be as follows.

Moreover, the procedure as shown herein is a standard, and if the effect of refining can be sufficiently obtained, it is allowed that this procedure is not always followed.

- 1) Transfer the concentrated solution into a separatory funnel (300 ml) while washing with 50 ml to 150 ml of hexane, add 5 ml of sulfuric acid, shake to mix gently and after allowing to stand remove the layer of sulfuric acid. Repeat this operation 3 or 4 times until the colour of sulfuric acid layer is light⁽¹⁴⁾.
- 2) Wash the hexane layer with 50 ml of hexane washing water, repeat the washing until the washings become almost neutral. After dehydrating with sodium sulfate, concentrate it to approximately 2 ml with a concentrator.
- 3) Put a plug of quartz glass wool into the bottom of the column-chromatograph tube in **6.3 d)**. Wash the inside of the tube with 10 ml of hexane and leave some hexane on the upper part of the quartz glass wool. Weigh out 3 g of silica gel in a beaker containing 10 ml of hexane, remove foam by stirring gently with glass rod and pour it in a column-chromatograph tube. After stabilizing the silica gel layer by running down hexane, put sodium sulfate approximately 10 mm in thickness on the layer and wash down sodium sulfate that adheres to the tube wall with several mls of hexane.

After running down 50 ml of hexane into a silica gel column, bring down the water level to the upper surface of sodium sulfate, transfer the solution prepared in 2) into the column gently, wash it several times with 1 ml of hexane. Bring down the water level to the surface of sodium sulfate. Set up a dropping cylindrical funnel containing 150 ml of hexane onto the top of the column chromatograph tube and run down hexane at the rate of approximately 2.5 ml/min (approximately one drop per second)⁽¹⁵⁾.

- 4) Concentrate the eluent to approximately 2 ml with a concentrator.

Notes ⁽¹⁴⁾ When adding the sulfuric acid, special care shall be taken regarding bumping of the solvents resulting in reactions between the sulfuric acid and organic substances. Start with several mls first, then add the acid gradually depending on the degree of coloration. Be sure to wear gloves, a mask, and other protective gear.

- (15) The elution conditions for dioxins in the operation of column chromatograph shall be confirmed by a fraction test by using the sample solution containing all dioxins such as extracted fluid of fly ash. The partition test shall be carried out at least every lot of column packing.

b) **Multilayer silica gel column chromatography** Operation of multilayer silica gel column chromatograph shall be according to the following procedures.

Moreover, the procedure as shown herein is a standard, and if the effect of refining can be sufficiently obtained, it is allowed that this procedure is not always followed.

- 1) Put a plug of quartz glass wool into the bottom of the column-chromatograph tube in **6.3 d** (inside diameter 15 mm) and pack in sequence, 0.9 g of silica gel, 3 g of silica gel impregnated with potassium hydroxide [2 % (mass fraction)], 0.9 g of silica gel, 4.5 g of silica gel impregnated with sulfuric acid [44 % (mass fraction)], 6 g of silica gel impregnated with sulfuric acid [22 % (mass fraction)], 0.9 g of silica gel, 3 g of silica gel impregnated with silver nitrate [10 % (mass fraction)] and 6 g of sodium sulfate into the tube⁽¹⁶⁾. This column is shown in figure 7.
- 2) After running down 50 ml of hexane, lower the water level to upper surface of sodium sulfate.
- 3) Transfer this solution into the column slowly, lower the water level to the upper end of the sodium sulfate.
- 4) Wash the container of the extracted solution with 1 ml of hexane and put the washings into the column, while rinsing its inside wall. Repeat this washing process 2 or 3 times.
- 5) Set up a dropping cylindrical funnel containing 120 ml of hexane onto the top of the column chromatograph tube and run the hexane down slowly at the rate of approximately 2.5 ml/min (approximately one drop per second)⁽¹⁵⁾.
- 6) Concentrate the eluent to approximately 2 ml with a concentrator. When the packed layers become dark in colour, repeat the procedures 1) to 6).

Note ⁽¹⁶⁾ Because the same effect as the treatment with sulfuric acid can only be obtained using the treatment with sulfuric acid silica gel for some samples, the column chromatograph tube for silica gel impregnated with sulfuric acid [22 % (mass fraction)] may be employed instead of the multilayer silica gel column-chromatograph tube. Silica gel impregnated with silver nitrate [10 % (mass fraction)] is effective for the treatment of the samples of specially high sulfur concentration.

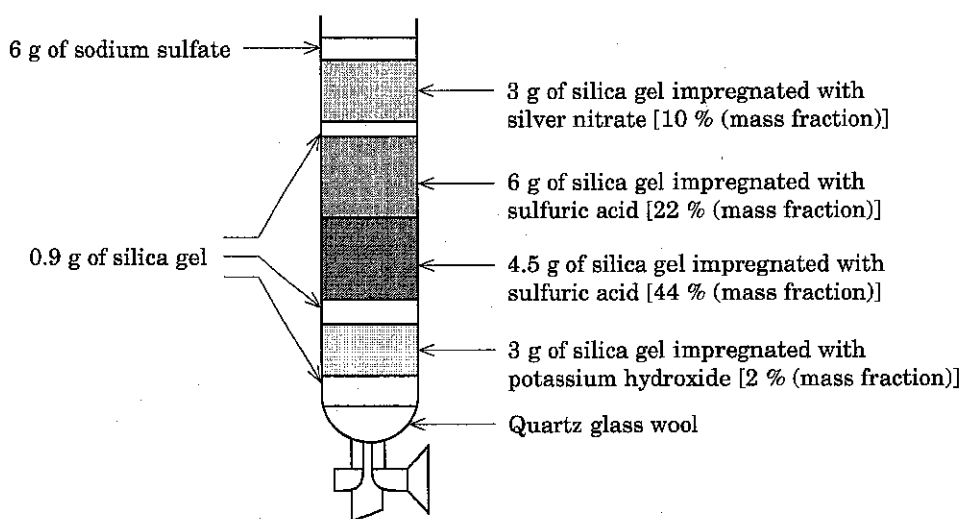


Figure 7 An example of multilayer silica gel column

6.4.5 Other refining operation The concentrated solution obtained in 6.4.4 shall be refined by means of either operation as shown in the following or their combination, and the concentrated solution for measuring PCDDs and PCDFs and for measuring DL-PCBs shall be prepared.

Moreover, the procedure as shown herein is a standard and if the effect of refining can be sufficiently obtained, it is allowed that this procedure is not always followed.

a) **Operation of alumina column chromatograph** For the operation of alumina column chromatograph, the procedure which is carried out so as to divide the concentrated solution into PCDDs and PCDFs measurement use and DL-PCBs measurement use shall be shown as follows. It is possible to carry out so as not to divide the concentrated solution depending on the operation conditions of identification and determination. The procedure in such a case is excluded from the rule.

1) **Operation of alumina column chromatograph for PCDDs and PCDFs**
The operation of alumina column chromatograph for PCDDs and PCDFs shall be according to following procedures.

- 1.1) Put a plug of quartz glass wool into the bottom of the column-chromatograph tube in 6.3 d). Wash the inside of the tube with 10 ml of hexane and leave some hexane on the upper part of quartz glass wool. Weigh out 10 g of alumina⁽¹⁷⁾ in a beaker containing 10 ml of hexane, remove foams by stirring gently with a glass rod and pour it into a tube of column-chromatograph. After stabilizing the alumina layer by running down hexane, add a layer of sodium sulfate of approximately 10 mm in thickness and wash down sodium sulfate adhering on the tube wall with several mls of hexane. After running down 50 ml of hexane, lower the water level to the upper surface of sodium sulfate.
- 1.2) After dividing the concentrated solution into two parts accurately, concentrate the one part of them up to approximately 0.5 ml by means of nitrogen current, and transfer it gently into the column of 1.1), wash it several

times with 1 ml of hexane, lower the water level to the surface of sodium sulfate. Run down 100 ml of hexane solution containing dichloromethane [2 % (volume fraction)] at the rate of approximately 2.5 ml/min (approximately one drop per second) and obtain the first fraction. This fraction shall be stored until the measurement is finished.

- 1.3) Further, run down 150 ml of hexane solution containing dichloromethane [50 % (volume fraction)] at the rate of approximately 2.5 ml/min and obtain the second fraction. PCDDs and PCDFs are included in this fraction⁽¹⁵⁾.
- 1.4) Concentrate the second fraction to approximately 2 ml with a concentrator.

Note ⁽¹⁷⁾ The activity of alumina can change significantly depending on the manufacturing lot and periods of storage after opening the seal. For alumina with lowered activity, 1,3,6,8-TeCDD and 1,3,6,8-TeCDF are eluted in the first fraction. Octachlorinated compounds may not be eluted into the second fraction with the specified volume of hexane solution containing dichloromethane [50 % (volume fraction)]. This shall be confirmed through fraction test using the extracted fluid of fly ash.

- 2) **Operation of alumina column chromatograph for dioxin like PCBs**
The operation of alumina column chromatograph for DL-PCBs shall be according to following procedures.

- 2.1) Put a plug of quartz glass wool at the bottom of the column-chromatograph tube in **6.3 d)**. Wash the inside of the tube with 10 ml of hexane and leave some hexane on the upper part of quartz glass wool. Weigh out 10 g of alumina⁽¹⁷⁾ in a beaker containing 10 ml of hexane, remove foams by stirring gently with glass rod and pour it in a tube of column-chromatograph. After stabilizing the alumina layer by running down hexane, add a layer of sodium sulfate approximately 10 mm in thickness and wash down sodium sulfate adhering on the tube wall with several mls of hexane. After flowing down 50 ml of hexane, lower the water level to the upper surface of sodium sulfate.
- 2.2) After dividing the concentrated solution into two parts accurately, concentrate the one part of them up to approximately 0.5 ml by means of nitrogen current and transfer it gently into a column of **2.1)**, wash it several times with 1 ml of hexane, lower the water level to the surface of sodium sulfate. Run down 40 ml of hexane solution at the rate of approximately 2.5 ml/min (approximately one drop per second) and elute aliphatic hydrocarbons⁽¹⁵⁾.
- 2.3) Run down 120 ml of hexane solution containing dichloromethane [5 % (volume fraction)] at the rate of approximately 2.5 ml/min (approximately one drop per second) and obtain the first fraction. DL-PCBs is contained in this fraction⁽¹⁵⁾.
- 2.4) Further, run down 160 ml of hexane solution containing dichloromethane [50 % (volume fraction)] at the rate of approximately 2.5 ml/min and obtain the second fraction. This fraction shall be stored until the measurement is finished.

- 2.5) Concentrate the first fraction to approximately 2 ml with a concentrator.
- b) **The operation of a high performance liquid chromatograph** The operation of high performance liquid chromatograph shall be in accordance with the following procedures. As the operation conditions shown hereafter may differ slightly depending on the apparatus or the column and others, differences shall be confirmed using the extracted fluid of fly ash by means of the fraction test.
- 1) The porous graphitized carbon column, of which direction for the flow through a mobile phase is changeable, is installed in a high performance liquid chromatograph with a passage changeable valve. The flow rate of eluant shall be set to 2 ml/min. Connect an absorbance detector and prepare it so that portions of the eluent can be taken from an exit of the detector.
 - 2) Run the toluene eluant toward the normal direction and wash the column thoroughly. Change the eluant to hexane and replace the toluene contained in the passage of the column and the apparatus with hexane. The degree of replacement to hexane should be judged by the change of the indicated value in the detector.
 - 3) Furthermore, concentrate the concentrated solution approximately up to 100 μ l by means of nitrogen current. Inject this solution into the high performance liquid chromatograph. Run the hexane for 4 min as the eluent. Take 8 ml aliquot of the eluent as the first fraction. PCBs except for DL-PCBs are contained in this fraction.
 - 4) Run the hexane solution containing dichloromethane [50 % (volume fraction)] for 20 min as the eluant. Take 40 ml aliquot of the eluent as the second fraction. Mono-ortho DL-PCBs are contained in this fraction.
 - 5) Run the hexane solution containing toluene [30 % (volume fraction)] for 20 min as the eluant. Take 40 ml aliquot of the eluent as the third fraction. Non-ortho DL-PCBs are contained in this fraction.
 - 6) Finally, heat an oven at 50 °C, reverse the direction of the mobile phase flow in the column and run toluene for 15 min. Take a 30 ml portion of the eluent as the fourth fraction. PCDDs and PCDFs are included in this fraction.
 - 7) Join the second fraction and the third fraction together, concentrate it approximately to 2 ml by the concentrator for measurement of DL-PCBs, and concentrate the fourth fraction similarly for measurement of PCDDs and PCDFs.
- c) **Operation of activated carbon column chromatograph** The operation of activated carbon column chromatograph shall be in accordance with the following procedure. The operation procedures shown herein are different depending on the column to be used, therefore it shall be confirmed to carry out the fraction test by using the extracted solution of fly ash, etc. beforehand.
- 1) Put a plug of quartz glass wool into the bottom of the column-chromatograph tube specified in 6.3 d). Layer sodium sulfate of approximately 10 mm in thickness, 1 g of silica gel impregnated with the porous graphitized carbon and sodium sulfate of approximately 10 mm in thickness again, one by

one. After washing it thoroughly with toluene run, replace the toluene in the column with hexane by running down hexane.

- 2) Furthermore, concentrate the concentrated solution approximately up to 100 μ l by means of nitrogen current. Load this solution to the column and make 50 ml of hexane run down at the rate of 2.5 ml/min.
 - 3) Run down 150 ml of hexane solution containing dichloromethane [25 % (volume fraction)] to obtain the first fraction. The mono-ortho compounds of DL-PCBs are contained in this fraction.
 - 4) Then, run down 200 ml of toluene to obtain the second fraction. Non-ortho compounds of PCDDs, PCDFs and DL-PCBs are contained in this fraction.
 - 5) Concentrate the first fraction and the second fraction respectively by using the concentrator approximately to 2 ml.
- d) **Operation of dimethyl sulfoxide (DMSO) partition treatment** The operation of dimethyl sulfoxide (DMSO) partition treatment shall be in accordance with the following procedures. This operation is to carry out for the purpose of removing low polar substance such as aliphatic hydrocarbon and can not divide into the measurement use of PCDDs and PCDFs and for the measurement use of DL-PCBs, therefore shall be carried out by combining with other refining operation.
- 1) Put 25 ml of hexane saturated DMSO in a separatory funnel, transfer the concentrated solution into this as washing with hexane, add 40 ml of hexane to the DMSO extracted solution of approximately 100 ml in total obtained by carrying out a shaking extraction four times, and wash.
 - 2) Put 75 ml of hexane and 100 ml of hexane washings in a separatory funnel, add approximately 100 ml of DMSO extracted solution obtained by the operation in 1), and carry out a shaking extraction three times. Obtain approximately 225 ml of hexane extracted solution.
 - 3) Put the hexane extracted solution obtained of approximately 225 ml in total in a separatory funnel, and carry out the washing by using 10 ml of potassium hydroxide aqueous solution (2 mol/L). Furthermore, wash with 25 ml of water twice, dehydrate with sodium sulfate, then concentrate it up to 2 ml by the concentrator.

6.4.6 Preparation of sample for measurement The internal standard substance for syringe spike is added to each concentrated solution for measurement of PCDDs and PCDFs and for measurement of DL-PCBs obtained according to the refining operation in 6.4.5 so as to be the same degree as the standard solution for preparation of calibration curve and 0.5 ml of nonane⁽¹⁸⁾ is added, and is made a definite liquid amount (20 μ l to 100 μ l) by means of nitrogen current⁽¹³⁾ again. This solution shall be taken as the sample for measurement of PCDDs and PCDFs and for measurement of DL-PCBs respectively.

Note ⁽¹⁸⁾ Toluene, decane or 2,2,4-trimethyl pentane may be used.

7 Identification and determination

7.1 Outline of the identification and the determination Identification and determination of dioxins shall be carried out using a gas chromatograph with a capillary column (GC) and a gas chromatograph coupled to a mass spectrometer with a double focussing mass spectrometer (MS). A resolution of 10 000 or over shall be required and 12 000 for some internal standard substances used. For the maintaining the measurement at high resolution of 10 000 or more, the determined ion monitoring (SIM method) is used for detection by the Lockmass mode. In this mode, the standard substances for correcting the mass and the sample for determination are led to an ion source simultaneously, and monitoring of the ions having near mass of the selected ions is carried out and the results are used to correct minute fluctuations of the mass. After dioxins are identified with their retention times and ionic strength ratios, determination of the amount shall be carried out using the peak area of the chromatogram by means of an internal standard method.

The detection limit of the GC/MS used in this Standard, though it is liable to vary depending on the apparatus or analytical conditions, is 0.1 pg for tetrachlorinated compounds and pentachlorinated compounds, 0.2 pg for hexachlorinated compounds and heptachlorinated compounds, 0.5 pg for octachlorinated compounds and 0.2 pg or under for DL-PCBs.

7.2 Reagents and apparatus

7.2.1 Reagents The reagents used for identification and determination shall be as follows.

- a) **Standard substance for the mass correction** The high boiling point of components such as perfluorokerosene (PFK), etc. shall be employed for a mass spectrometry.
- b) **Standard substance** The standard substances used for identification and determination of dioxins using the internal standard method are given in table 3.
- c) **Internal standard substance** Internal standard substances used for sampling spike, cleanup spike and syringe spike shall be used [refer to 5.3 and note (7)].
- d) **Standard solution for preparing the calibration curve** This solution shall be prepared by mixing the standard substance of b) with the internal standard substance for the cleanup spike and syringe spike, and by diluting with nonane⁽¹⁸⁾ into five steps ranging from low concentration of approximately 3-fold of detection limit, in the range of the quantitative analysis for GC/MS. (Refer to annex 2 for an example for the preparation of the standard solutions of the calibration curve).

Table 3 Standard substances for dioxins

		Compound
PCDDs	Tetrachlorinated compounds	2,3,7,8-TeCDD
	Pentachlorinated compounds	1,2,3,7,8-PeCDD
	Hexachlorinated compounds	1,2,3,4,7,8-HxCDD
		1,2,3,6,7,8-HxCDD
		1,2,3,7,8,9-HxCDD
	Heptachlorinated compounds	1,2,3,4,6,7,8-HpCDD
	Octachlorinated compounds	1,2,3,4,6,7,8,9-OCDD
PCDFs	Tetrachlorinated compounds	2,3,7,8-TeCDF
	Pentachlorinated compounds	1,2,3,7,8-PeCDF
		2,3,4,7,8-PeCDF
	Hexachlorinated compounds	1,2,3,4,7,8-HxCDF
		1,2,3,6,7,8-HxCDF
		1,2,3,7,8,9-HxCDF
		2,3,4,6,7,8-HxCDF
	Heptachlorinated compounds	1,2,3,4,6,7,8-HpCDF
		1,2,3,4,7,8,9-HpCDF
	Octachlorinated compounds	1,2,3,4,6,7,8,9-OCDF
DL-PCBs*	Tetrachlorinated compounds	3,3',4,4'-TeCB (#77)
		3,4,4',5-TeCB (#81)
	Pentachlorinated compounds	2,3,3',4,4'-PeCB (#105)
		2,3,4,4',5-PeCB (#114)
		2,3',4,4',5-PeCB (#118)
		2',3,4,4',5-PeCB (#123)
		3,3',4,4',5-PeCB (#126)
	Hexachlorinated compounds	2,3,3',4,4',5-HxCB (#156)
		2,3,3',4,4',5'-HxCB (#157)
		2,3',4,4',5,5'-HxCB (#167)
		3,3',4,4',5,5'-HxCB (#169)
		Compound
	Heptachlorinated compounds	2,3,3',4,4,4',5'-HpCB (#189)

Note * Numerical values in parentheses show the number of IUPAC.

7.2.2 Gas chromatograph mass spectrometer (GC/MS) Gas chromatograph coupled to a mass spectrometer used for the identification and the determination of dioxins shall be as follows.

a) Gas chromatograph (GC)

- Sample injection part** This part shall be operational at 250 °C to 280 °C with splitless, on column injection mode or a large volume injection system (programmed temperature vapourization injection system, etc.)⁽¹⁹⁾.

Note ⁽¹⁹⁾ In the case of a large volume injection system, there is the possibility that the determination is influenced caused by the dechlorination of dioxins. For example, when OCDD is higher concentration as compared with the concentration of HpCDDs, the determination of HpCDDs is considered to be not accurate, therefore care shall be taken.

- 2) **Column** Capillary column made of fused silica that shall be 0.1 mm to 0.52 mm in inside diameter and 25 m to 60 m in length.

In the determination of PCDDs and PCDFs, a column shall be used, in which isomers substituted by chlorine at 2,3,7,8-positions are isolated as much as possible and the elution sequence for all compounds are proved on the chromatogram. The column, in which all isomers substituted by chlorine at 2,3,7,8-positions can be completely separated from other isomers, has not been reported, therefore it is recommended that all isomers substituted by chlorine at 2,3,7,8-positions can be singly determined by using two kinds or over of columns of different elution sequence together. When there are isomers substituted by chlorine at 2,3,7,8-positions which can not be singly determined, the influence of overlapped isomer can not be ignored and the measurement results are much influenced in some case, therefore care shall be taken [see a) of 8.1].

In the determination of DL-PCBs, a column shall be used, in which 12 kinds of DL-PCBs can be isolated from other PCB compound as much as possible and the elution sequence for all PCB compounds from tetra-chlorination to deca-chlorination are proved on the chromatogram.

Information : There are BPX-DXN (SGE), CPS-1 (Quadrex), CP-Sil 88 (Chrompack), DB-5 (J&W), DB-17 (J&W), DB-210 (J&W), DB-225 (J&W), OV-17 (Quadrex), RH-12ms (Inventx), SP-2331 (Supelco), etc. as the columns in which the elution sequence of PCDDs and PCDFs are reported, and there are DB-5ms (J&W), HT8 (SGE), RH-12ms (Inventx), etc. as the columns in which the elution sequence of PCB are reported. These commodities are mentioned as available commonly, however, they are not the case to recommend.

- 3) **Carrier gas** High purity helium with a purity of 99.999 % (volume fraction) or more.
- 4) **Thermostatic column tank** The temperature control ranges shall be 50 °C to 350 °C and it shall be equipped with a temperature rising program to provide the optimum separation conditions for the substances being measured.

b) **Mass spectrometer (MS)**

- 1) **Method** Double-focusing
- 2) **Resolution** 10 000 or over. However, when $^{13}\text{C}_{12}$ -OCDF is used as the internal standard substance, approximately 12 000 may be required depending on the selection of the capillary column.
- 3) **Ion detection method** Selected ion detection (SIM) method by Lockmass mode using the standard substance for the mass correction
- 4) **Ionization mode** Electron bombardment ionization (EI) method
- 5) **Ionic source temperature** 250 °C to 300 °C
- 6) **Ionization current** 0.5 mA to 1 mA
- 7) **Electron acceleration voltage** 30 V to 70 V

- 8) **Maximum ion acceleration voltage** 5 kV to 10 kV

7.3 Measurement

7.3.1 Set up to measuring conditions for a gas chromatograph mass spectrometer Set up of measuring conditions for a gas chromatograph mass spectrometer shall be as follows.

- a) **Gas chromatograph (GC)** The operational conditions of a gas chromatograph for PCDDs and PCDFs and DL-PCBs shall be as follows.
- 1) **PCDDs and PCDFs** In the determination of PCDDs and PCDFs, the conditions of gas chromatograph shall be set up in order to obtain good separation of the peaks of isomers substituted by chlorine at 2,3,7,8-positions from that of the other isomers on a chromatogram, to ensure retention time of each chlorine compound is within an appropriate range and to obtain a stable response. The separation condition of each compound under the condition of setting up shall be confirmed beforehand by measuring the sample such as the extracted solution of fly ash, etc. An example is shown in attached tables 1 to 3.
 - 2) **DL-PCBs** In the determination of DL-PCBs, the condition of gas chromatograph shall be set up in order to obtain good separation of the peaks of DL-PCBs from that of the other isomers on a chromatogram, to ensure retention time of each chlorine compound is within an appropriate range and to obtain a stable response. The separation condition of each compound under the condition of setting up shall be confirmed beforehand, by measuring the sample such as the extracted solution of fly ash, etc. An example is shown in attached tables 4 and 5.

Information : The conditions shown here are given for information.

- b) **Mass spectrometer (MS)** The mass spectrometer shall be set up in order to satisfy the following conditions.
- 1) **Resolution** Resolution shall be 10 000 or over. When $^{13}\text{C}_{12}$ -OCDF is employed as an internal standard substance, depending on the selection of a gas chromatograph column, approximately 12 000 shall be required.
 - 2) **Ion detection method** Selected ion detection (SIM) method by Lockmass mode that the standard substances for the mass correction are used.
 - 3) **Measured mass number/electric charge number (m/z)** Mass number/electric charge number (m/z)⁽²⁰⁾ of two or more selected ions and those for the Lockmass mode shall be set for every chlorine compound in the samples and the internal standard substances. An example of set mass number/electric charge number (m/z) of PCDDs and PCDFs is given in table 4, and that of DL-PCBs is given in table 5 respectively.

Note ⁽²⁰⁾ While the peak widths obtained with a capillary column are approximately 5 s to 10 s, in order to obtain many data points enough to one peak, even if the peak of single component on chromatogram is the peak of narrowest width, the sampling period of selected ion detection shall be set up so that the measuring points

constituting the peak are seven or more points. The number of monitor channels which can be set up at the one measurement depends on the sensitivity required and therefore shall be set after sufficient check.

As considering the retention time of each peak on the chromatogram, the measurement by grouping the retention time with time division mode is acceptable. In this case, however, appropriate conditions shall be set so that a relevant internal standard substance peak appears for each group.

7.3.2 Adjustment of the mass spectrometer Adjustment of the mass spectrometer shall be carried out with a mass correction program by introducing the standard substance for mass correction, after the conditions for the required items are set while the apparatus is in operation. The mass scale, resolution (at least 10 000), etc. shall be calibrated to a fixed value depending on the measurement purpose. In particular, the resolution shall be adjusted to at least 10 000 in all parts of the mass ranges measured. Usually, it shall be carried out before the start of measurements. The results of the mass correction shall be stored.

7.3.3 Operation of SIM detection The operation of the SIM detection shall be carried out as follows.

- a) Set the required GC/MS conditions.
- b) After a standard substance for mass correction is introduced and the response of the monitor channel becomes stable, a sample is measured.
- c) The chromatogram shall be recorded on the mass number of each fixed chlorinated compound.
- d) After the measurement, the monitor channel for internal standard substance for mass correction, existence of the interferences, the separation of isomers substituted by chlorine at 2,3,7,8-positions shall be confirmed for each sample, prior to the data processing operations⁽²¹⁾.

Note ⁽²¹⁾ If a chromatogram in the monitor channel of the standard substance for mass correction shows the signal variation of not less than $\pm 20\%$ in the appearance time of the object component for the determination, these compounds shall not be determined. The main cause for this may be an insufficient pretreatment of the sample. To counteract this, sample pretreatment shall be thoroughly repeated and Lockmass fluctuations shall be minimized.

Table 4 An example of setting mass number/electric charge number (monitor ion)* for PCDDs and PCDFs measurement

	Substituent compounds by chlorine	M^+	$(M+2)^+$	$(M+4)^+$
Target substances	TeCDDs	319.896 5	321.893 6	
	PeCDDs	353.857 6	355.854 6	357.851 7**
	HxCDDs	387.818 6	389.815 6	391.812 7**
	HpCDDs		423.776 7	425.773 7
	OCDD		457.737 7	459.734 8
	TeCDFs	303.901 6	305.898 7	
	PeCDFs		339.859 7	341.856 8
	HxCDFs		373.820 7	375.817 8
	HpCDFs		407.781 8	409.778 8
	OCDF	439.745 7	441.742 8	443.739 8
Internal standard substances	$^{13}\text{C}_{12}$ -TeCDDs	331.936 8	333.933 9	
	$^{37}\text{Cl}_4$ -TeCDDs	327.884 7		
	$^{13}\text{C}_{12}$ -PeCDDs	365.897 8	367.894 9	369.891 9
	$^{13}\text{C}_{12}$ -HxCDDs	399.858 9	401.855 9	403.853 0
	$^{13}\text{C}_{12}$ -HpCDDs		435.816 9	437.814 0
	$^{13}\text{C}_{12}$ -OCDD		469.778 0	471.775 0
	$^{13}\text{C}_{12}$ -TeCDFs	315.941 9	317.938 9	
	$^{13}\text{C}_{12}$ -PeCDFs		351.900 0	353.897 0
	$^{13}\text{C}_{12}$ -HxCDFs		385.861 0	387.858 0
	$^{13}\text{C}_{12}$ -HpCDFs		419.822 0	421.819 1
	$^{13}\text{C}_{12}$ -OCDF	451.786 0	453.783 0	455.780 1
Standard substance for mass correction (PFK)		330.979 2 (for determination of 4,5-chlorinated compounds)		
		380.976 0 (for determination of 5,6-chlorinated compounds)		
		430.972 9 (for determination of 7,8-chlorinated compounds)		
		442.972 9 (for determination of 7,8-chlorinated compounds)		

Notes * Mass numbers are calculated on the basis of "the IUPAC data, Element by element review of their atomic weight, Pure Appl Chem., 56, [6] p. 695 to 768 (1984)".

** There is a possibility of interference by PCBs.

Table 5 An example of setting mass number/electric charge number (monitor ion)* for the DL-PCBs measurement

	Substituent compounds by chlorine	M^+	$(M+2)^+$	$(M+4)^+$
Target substances	TeCBs	289.922 4	291.919 4	293.916 5
	PeCBs	323.883 4	325.880 4	327.877 5
	HxCBs	357.844 4	359.841 5	361.838 5
	HpCBs	391.805 4	393.802 5	395.799 5
Internal standard substances	$^{13}\text{C}_{12}$ -TeCBs	301.962 6	303.959 7	305.956 7
	$^{13}\text{C}_{12}$ -PeCBs	335.923 7	337.920 7	339.917 8
	$^{13}\text{C}_{12}$ -HxCBs	369.884 7	371.881 7	373.878 8
	$^{13}\text{C}_{12}$ -HpCBs	403.845 7	405.842 8	407.839 8
Standard substance for mass correction (PFK)			304.982 4	
			330.979 2	
			380.976 0	

Note * Mass numbers are calculated on the basis of "the IUPAC data, Element by element review of their atomic weight, Pure Appl Chem., 56, [6] p. 695 to 768 (1984)".

7.3.4 Preparation of the calibration curve Preparation of the calibration curve shall be as follows.

- a) **Measurements of the standard solution** Inject the standard solution for a calibration curve into the GC/MS at least three times for one concentration, carry out the operation of SIM detection in 7.3.3, and obtain a total of 15 or more pieces of data in all concentration areas.
- b) **Confirmation for the strength ratio of the peak area** Using the obtained chromatogram, calculate the strength ratio of the peak area for each standard substance to that for the corresponding two mass number ions, and conform that the ratio is almost in agreement with the ion strength ratio, estimated from the isotope abundance ratio of chlorine atom within $\pm 15\%$ (refer to 7.4.1 table 6).
- c) **Calculation of the relative response factor** Calculation of the relative response factor shall be as follows.
 - 1) Obtain the peak area for each standard substance and internal standard substance. Prepare the calibration curve using the ratio of peak area for each standard substance to that for the corresponding cleanup spike internal standard substance and the concentration ratio between the standard substance and that of the internal standard substance in the standard solution which has been injected and confirm that the calibration curve is the straight line passing the origin (refer to annex 2 for an example for the standard substances of the measuring target and the corresponding internal standard substances).

RRF_{es} shall be obtained for each determination by using formula (3), and the obtained data of not less than 15 points in all region of concentration shall be averaged. In this case, the coefficient of variation of data shall be minified to 5 % as the aim as much as possible and the compounds exceeding

10 % in coefficient of variation shall not be found in presence. When the coefficient of variation exceeds 10 %, the condition of GC/MS shall be confirmed and if necessary, the calibration curve shall be prepared again by taking the measure so as to adjust again or to narrow the range of determination in the range having linearity, etc.

$$RRF_{cs} = \frac{Q_{cs}}{Q_s} \times \frac{A_s}{A_{cs}} \dots\dots\dots (3)$$

where, RRF_{cs} : relative response factor of the measuring target component against the internal standard substance for cleanup spike

Q_{cs} : amount of the internal standard substance for cleanup spike in the standard solution (pg)

Q_s : amount of the measuring target substance in the standard solution (pg)

A_s : peak area of the measuring target substance in the standard solution⁽²²⁾

A_{cs} : peak area of the internal standard substance for cleanup spike in the standard solution⁽²²⁾

Note ⁽²²⁾ The peak area used herein shall be any one of the peak area of one side of measurement channel, the total value of peak area of both measurement channels or the mean value of peak area of both measurement channels, and the same one shall be used in all measurement up to the measurement of sample.

- 2) In the same manner as above, the relative response factor (RRF_{rs}) of the internal standard substances for cleanup spike against those for syringe spike shall be calculated with formula (4), and the relative response factor (RRF_{ss}) of the internal standard substances for sampling spike against those for cleanup spike shall be calculated with formula (5). (Refer to annex 2 for an example of correspondence of the internal standard substances for cleanup spike to those for syringe spike and the internal standard substances for sampling spike to those for cleanup spike).

$$RRF_{rs} = \frac{Q_{rs}}{Q_{cs}} \times \frac{A_{cs}}{A_{rs}} \dots\dots\dots (4)$$

where, RRF_{rs} : relative response factor of the internal standard substance for cleanup spike against the internal standard substance for syringe spike

Q_{rs} : amount of the internal standard substance for syringe spike in the standard solution (pg)

Q_{cs} : amount of the internal standard substance for cleanup spike in the standard solution (pg)

A_{cs} : peak area of the internal standard substance for cleanup spike in the standard solution⁽²²⁾

A_{rs} : peak area of the internal standard substance for syringe spike in the standard solution⁽²²⁾

$$RRF_{ss} = \frac{Q_{cs}}{Q_{ss}} \times \frac{A_{ss}}{A_{cs}} \dots\dots\dots (5)$$

where, RRF_{ss} : relative response factor of the internal standard substance for sampling spike against the internal standard substance for cleanup spike

Q_{cs} : amount of the internal standard substance for cleanup spike in the standard solution (pg)

Q_{ss} : amount of the internal standard substance for sampling spike in the standard solution (pg)

A_{ss} : peak area of the internal standard substance for sampling spike in the standard solution⁽²²⁾

A_{cs} : peak area of the internal standard substance for cleanup spike in the standard solution⁽²²⁾

7.3.5 Measurement of the sample Measurement of the sample prepared in clause 6 shall be as follows.

- a) **Confirmation of the calibration curve** One or more in the standard solutions for preparing calibration curve shall be selected at a certain definite period (one or more per day) and measured according to the SIM detection in 7.3.3 and the relative response factor (RRF_{cs}) of the internal standard substances for cleanup spike corresponding for that of each compounds shall be obtained according to the same manner as in 7.3.4. The relative response factor (RRF_{rs}) of the internal standard substances for the cleanup spike against the internal standard substance for the syringe spike shall be obtained.

If these relative response factors is within $\pm 10 \%$ for RRF_{cs} and within $\pm 20 \%$ for RRF_{rs} against the relative response factors (RRF_{cs} and RRF_{rs}) at preparation of calibration curve obtained in 7.3.4, measurement shall be carried out using the relative response factor obtained in 7.3.4. In the case of deviating from this range, eliminate the cause for this and measure again or prepare the calibration curve again.

Furthermore, for the retention time, the variation shall be investigated and when the retention time varies by not less than $\pm 5 \%$ per day and the relative retention to internal standard substance varies by not less than $\pm 2 \%$, the cause shall be eliminated and the remeasurement for a series of samples carried out just before shall be carried out.

- b) **Measurement of the sample** The sample prepared in clause 6 shall be measured according to the SIM detection in 7.3.3 and the chromatogram shall be obtained for the mass number of each chlorinated compound.

7.4 Identification and determination

7.4.1 Detection of the peaks The detection of peaks on a chromatogram obtained in 7.3.5 shall be carried out as follows.

- a) **Confirmation of the internal standard substance for the syringe spike** It shall be confirmed that the peak area of the internal standard substance for syringe

spike in the sample prepared in 6.4.6 is beyond 70 % against that in the standard solution. When it is out of the above range, measurement shall be carried out again after finding and eliminating the cause for the fluctuation.

- b) **Detection of the peaks** For the peak of which peak height (S) is at least 3-fold against the noise width (N) of the base line on the chromatogram, i.e. for a peak of $S/N = 3$ or more in peak height, the following identification and determination shall be carried out.

Here, the noise width (N) and peak height (S) are obtained generally according to the following procedures. Measure noises close to a peak, (within approximately 10-fold range for the half width of the peak), and take, as the noise width (N), twice of the standard deviation of the measured values. Alternatively, the difference between the maximum value and the minimum value of the noise height is multiplied by 2/5 to obtain the noise width (N) because it is known empirically that the width between the maximum and the minimum of the noise height is equivalent to approximately 5 times its standard deviation. On the other hand, take the median of the noise as a base line and obtain the height of the top peak from this noise line. This height is taken as the peak height (S).

However, because the noise cannot be measured if the base line of the chromatogram obtained is not higher than the zero point of the apparatus, the base line shall be ascertained prior to the measurement and, as required, the offset, etc. shall be adjusted as appropriate.

- c) **Calculation of the peak area** The peak area shall be calculated using the peak detected in b).

7.4.2 Identification of dioxins The Identification of the sample shall be as follows.

- a) **Identification of dioxins** When the peak area ratio on the chromatogram for at least two monitored ions is almost equal to that of the standard substance and the ion strength ratio is within $\pm 15 \%$, ($\pm 25 \%$ for the concentrations which are not more than 3-fold of the minimum limits of detection), against that of values estimated with the isotopic abundance of chlorine atoms given in table 6, the peak shall be taken as dioxins or PCBs. The identification of compounds without standard substance shall be carried out by referring to the elution sequence of columns used for analysis.
- b) **Identification of isomers substituted by chlorine at 2,3,7,8-positions** The compounds substituted by chlorine at 2,3,7,8-positions contained in PCDDs and PCDFs identified shall be identified by the retention time of the peak on the chromatogram being almost equal to those of the standard substances and the relative retention time against the corresponding internal standard substance coincides with that of the standard substance.
- c) **Identification of DL-PCBs** The DL-PCBs in identified PCBs of which the retention time of the peak on the chromatogram is almost equal to those of the standard substances is identified by the fact that relative retention time against the corresponding internal standard substance coincides with that of standard substance.

Table 6 Ionic strength ratios estimated with the isotopic abundance of chlorine atoms

	<i>M</i>	<i>M</i> +2	<i>M</i> +4	<i>M</i> +6	<i>M</i> +8	<i>M</i> +10	<i>M</i> +12	<i>M</i> +14
TeCDDs	77.43	100.00	48.74	10.72	0.94	0.01		
PeCDDs	62.06	100.00	64.69	21.08	3.50	0.25		
HxCDDs	51.79	100.00	80.66	34.85	8.54	1.14	0.07	
HpCDDs	44.43	100.00	96.64	52.03	16.89	3.32	0.37	0.02
OCDD	34.54	88.80	100.00	64.48	26.07	6.78	1.11	0.11
TeCDFs	77.55	100.00	48.61	10.64	0.92			
PeCDFs	62.14	100.00	64.57	20.98	3.46	0.24		
HxCDFs	51.84	100.00	80.54	34.72	8.48	1.12	0.07	
HpCDFs	44.47	100.00	96.52	51.88	16.80	3.29	0.37	0.02
OCDF	34.61	88.89	100.00	64.39	25.98	6.74	1.10	0.11
TeCBs	76.67	100.00	49.11	10.83	0.93			
PeCBs	61.42	100.00	65.29	21.43	3.56			
HxCBs	51.22	100.00	81.48	35.51	8.75	1.17		
HpCBs	43.93	100.00	97.67	53.09	17.38	3.43		

Remarks 1 'M' refers to the isotope with the minimum mass number.

2 Ionic strength ratios refer to the values for homologue when the value for the isomer having the strongest intensity is set as 100 %.

7.4.3 Determination of dioxins

- a) **Determination of each compound** The amount (Q_i) of the isomers substituted by chlorine at 2,3,7,8-positions in dioxins identified or the DL-PCBs in the total volume of the extracted solution shall be calculated using formula (6) by using the internal standard method based on the added amount of the corresponding internal standard substance for cleanup spike. The same procedures shall be employed for any other compounds. (Refer to annex 2 for an example of the substances for measuring target components, the standard substances and the corresponding internal standard substance in the cleanup).

$$Q_i = \frac{A_i}{A_{csi}} \times \frac{Q_{csi}}{RRF_{cs}} \dots\dots\dots (6)$$

where, Q_i : amount of the compound in the total volumes of the extracted solution (ng)

A_i : peak area of the compound on the chromatogram⁽²³⁾

A_{csi} : peak area of corresponding internal standard substance for cleanup spike⁽²²⁾

Q_{csi} : added amount of corresponding internal standard substance for cleanup spike (ng)⁽²³⁾

RRF_{cs} : relative response factor against the corresponding internal standard substance for cleanup spike⁽²⁴⁾

Notes ⁽²³⁾ When, after extracting a sample, an aliquot of the solution is taken and the internal standard substance is added, the corrections shall be carried out.

- (24) The determination of the compounds other than isomers substituted by chlorine at 2,3,7,8-positions and DL-PCBs, the average value of the relative response factor between the standard substance used for each chlorine compound and the standard substance for cleanup shall be used.

- b) **Calculation of the concentration** The concentrations in the sample shall be calculated using formula (7) by using the amount of each compound obtained. Unless otherwise specified, the numerical values shall be rounded off to two significant figures in accordance with **JIS Z 8401**.

$$C_i = (Q_i - Q_t) \times \frac{1}{V_{SD}} \quad \text{..... (7)}$$

- where, C_i : concentration of the compound in the sample (0 °C, 101.32 kPa) (ng/m³)
 Q_i : amount of the compound in the total volumes of the extracted solution (ng)
 Q_t : amount of the compound in the blank test (ng)
 V_{SD} : sampling volume of the sample gas under 0 °C and 101.32 kPa (m³)

Remarks : When correction with the concentration of oxygen is needed, convert the concentration obtained by actual measurement into the designated concentration with the formula (8) and use this as the concentration.

$$C = \frac{21 - O_n}{21 - O_s} \times C_s \quad \text{..... (8)}$$

- where, C : concentration in the oxygen concentration O_n (0 °C, 101.32 kPa) (ng/m³)
 O_n : oxygen concentration to be converted (%)
 O_s : oxygen concentration in the exhaust gas⁽²⁵⁾ (%)
 C_s : concentration obtained by actual measurement in the exhaust gas (0 °C, 101.32 kPa) (ng/m³)

Note ⁽²⁵⁾ If oxygen concentration in exhaust gas is beyond 20 %, take $O_s = 20$.

7.5 The minimum limit of detection and the minimum limit of determination

7.5.1 The minimum limit of detection and minimum limit of determination for the apparatus Measure the standard solution for preparing the calibration curve of the minimum concentration (containing each standard substance such as 0.1 pg to 0.5 pg for tetra- and penta-chlorinated compounds, 0.2 pg to 1.0 pg for hexa- and hepta-chlorinated compounds, 0.5 pg to 2.5 pg for octa-chlorinated compounds and 0.2 pg to 1.0 pg for DL-PCBs) with GC/MS, and determine each compound. Repeat this operation 5 times or more and calculate the standard deviation using formula (9) by using the measured values obtained. Take threefold of the standard deviation as the minimum limit of detection for the apparatus and tenfold of the standard deviation as the minimum limit for the determination. Herein, calculate

the standard deviation so as not to carry out the rounding of measured value, make the minimum limit of detection obtained to one place of significant figure and round off the minimum limit of determination up to the same place of the minimum limit of detection.

When the minimum limits of detection for the apparatus are larger than 0.1 pg for tetra- and penta-chlorinated compounds, 0.2 pg for hexa- and hepta-chlorinated compounds, 0.5 pg for octa-chlorinated compounds and 0.2 pg for DL-PCBs, check the equipment and the apparatus, and adjust them to smaller than these values.

As the minimum limit of detection and the minimum limit of the determination for the apparatus are liable to vary depending on the conditions, etc. of the GC/MS used, check it at regular intervals and control the apparatus so that it always gives satisfactory results. It shall definitely be confirmed when the GC/MS is used and conditions of measurement have been changed.

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}} \dots\dots\dots (9)$$

where, s : standard deviation
 x_i : individual data of measurements (pg)
 $\bar{x}$: average of data (pg)
 n : number of measurements

7.5.2 The minimum limit of detection and the minimum limit of determination for the measuring method Add the amount of the standard substance, calculated using formula (10), into the extracted solution obtained from the same volume of the absorption solution, extracted solution from adsorbent and filter used in the measurement, carry out pretreatment, measurement, identification and determination of the sample with GC/MS. Repeat this operation 5 times or more, calculate the standard deviation of the data obtained using formula (9). Take threefold of the standard deviation as the minimum limit of detection for the measuring method and tenfold of the above standard deviation as the minimum limit of the determination for the measuring method. Herein, calculate the standard deviation so as not to carry out the rounding of measured value, make the minimum limit of detection obtained to one place of significant figure and round the minimum limit of determination up to the same place of the minimum limit of detection.

As the minimum limit of detection and the minimum limit of determination for the measuring method are liable to vary depending on the conditions of pretreatment and measurement used, check them at regular intervals and control them so that satisfactory results are always obtained. It shall definitely be confirmed in the case where pretreatment operations and conditions of measurement have been changed.

$$Q = QL' \times \frac{v}{v_i} \dots\dots\dots (10)$$

where, Q : added amount of the standard substances (pg)
 QL' : minimum limit of determination for the apparatus (pg)
 v : volumes of the sample solution (μl)

v_i : injecting volume into GC/MS (μl)

7.5.3 The minimum limit of detection and the minimum limit of determination for the sample gas As the limits of the sample gas are liable to vary depending on the sampling volume, they shall be obtained for every sample by using formulae (11) and (12).

$$C_{DL} = \frac{DL}{1\,000} \times \frac{v}{v_i} \times \frac{V_E}{V'_E} \times \frac{1}{V} \dots\dots\dots (11)$$

$$C_{QL} = \frac{QL}{1\,000} \times \frac{v}{v_i} \times \frac{V_E}{V'_E} \times \frac{1}{V} \dots\dots\dots (12)$$

where, C_{DL} : minimum limit of detection for the sample gas (0 °C, 101.32 kPa) (ng/m^3)

C_{QL} : minimum limit of determination for the sample gas (0 °C, 101.32 kPa) (ng/m^3)

DL : minimum limit of detection for measuring method (pg)

QL : minimum limit of determination for measuring method (pg)

v_i : injecting volume into GC/MS (μl)

v : volumes of the sample solution (μl)

V : sampling volume of sample gas (0 °C, 101.32 kPa) (m^3)

V_E : volume of extracted solution (ml)

V'_E : aliquot volume of extracted solution (ml)

7.5.4 Confirmation of the minimum limit of detection When measuring the actual sample, if the peak can not be detected for the isomers substituted by chlorine at 2,3,7,8-positions and DL-PCBs, the minimum limit of detection shall be obtained on the chromatogram as given below. It shall be confirmed that the concentration in the sample gas calculated from this value is not more than the minimum limit of detection in the sample gas obtained in 7.5.3. When this value exceeds the minimum limit of detection in sample gas, it shall be confirmed whether a problem is found in presence in pretreatment operation, measurement operation or not, then the measurement shall be carried out after eliminating the cause and the concentration in sample gas calculated from the minimum limit of detection at the time of measurement of sample shall be made so as to be not more than 1/30 of the concentration to be evaluated.

- Measure the noise width of the base line close to the peak of the compounds.
- Estimate the peak area which is equivalent to the height corresponding to 3-fold of the noise width using the chromatogram of standard solution.
- Calculate the amount corresponding to the area by using the peak area obtained, and take it as the minimum limit of detection at the time of measurement of sample.

7.6 Confirmation of the recovery ratio

7.6.1 Calculation of the recovery ratio for the cleanup spike The recovery ratio of the cleanup spike shall be calculated from formula (13) by using the ratio between the peak area of internal standard substance for the cleanup spike against that of internal standard substance for syringe spike and by using the corresponding relative response factor (RRF_{rs}) to confirm the recovery ratio of cleanup spike.

If the recovery ratio of cleanup spike is out of the range between 50 % and 120 %, pretreatment and measurement shall be carried out again.

$$R_c = \frac{A_{csi}}{A_{rsi}} \times \frac{Q_{rsi}}{RRF_{rs}} \times \frac{100}{Q_{csi}} \dots\dots\dots (13)$$

- where, R_c : recovery ratio of cleanup spike (%)
- A_{csi} : peak area of internal standard substance for cleanup spike⁽²²⁾
- A_{rsi} : peak area of corresponding internal standard substance for syringe spike⁽²²⁾
- Q_{rsi} : added amount of corresponding internal standard substance for syringe spike (ng)
- RRF_{rs} : relative response factor of corresponding internal standard substance for syringe spike
- Q_{csi} : added amount of internal standard substance for cleanup spike (ng)⁽²⁶⁾

Note ⁽²⁶⁾ After adding the internal standard substance, it shall be corrected for aliquot volume and partition degree.

7.6.2 Calculation of the recovery ratio for the sampling spike The recovery ratio of sampling spike shall be calculated from formula (14) by using the ratio between the peak area of internal standard substance for sampling spike against that of the internal standard substance for the cleanup spike, and the corresponding relative response factor (RRF_{ss}) to confirm the recovery ratio of sampling spike.

If the recovery ratio of sampling spike is out of the range between 70 % and 130 %, eliminate the cause, and sampling and measuring shall be carried out again.

$$R_s = \frac{A_{ssi}}{A_{csi}} \times \frac{Q_{csi}}{RRF_{ss}} \times \frac{100}{Q_{ssi}} \dots\dots\dots (14)$$

- where, R_s : recovery ratio of sampling spike (%)
- A_{ssi} : peak area of internal standard substance for sampling spike⁽²²⁾
- A_{csi} : peak area of corresponded internal standard substance for cleanup spike⁽²²⁾
- Q_{csi} : added amount of corresponded internal standard substance for cleanup spike (ng)
- RRF_{ss} : relative response factor of corresponding internal standard substance for cleanup spike

Q_{ssi} : added amount of internal standard substance for sampling spike (ng)⁽²⁶⁾

8 Report of the results

8.1 Methods of expression for analytical results Unless otherwise specified, the expression method of the analytical results for dioxins shall be as follows.

- a) **PCDDs and PCDFs** The concentrations of each isomer substituted by chlorine at 2,3,7,8-positions, those of the homologous series of tetrachlorinated compounds to octa-chlorinated compounds (TeCDDs to OCDD and TeCDFs to OCDF) and their total concentration shall be recorded in the analytical results of PCDDs and PCDFs.

If the concentrations of each compounds are not less than the minimum limit of determination for the sample gas, concentrations shall be recorded as they are. If the concentrations are not less than the minimum limit of detection and less than the minimum limit of determination, it shall be recorded by such understandable expression method (e.g. value in parentheses or in other column) that the values are not guaranteed the same analytical precision as those of not less than the minimum limit of determination. If the values are less than the minimum limit of detection for the sample gas, they shall be recorded by such a method that the values are understood to be under the minimum limit of detection.

For the isomers substituted by chlorine at 2,3,7,8-positions which are not singly determined, the designation of isomer which is overlapped in the column of isomers substituted by chlorine at 2,3,7,8-positions in the result table shall be clearly mentioned so as to prove that it is not singly determined. For example, when 1,2,3,4,8-PeCDF is overlapped on 1,2,3,7,8-PeCDF, it shall be mentioned in the column of 1,2,3,7,8-PeCDF as "1,2,3,7,8 + 1,2,3,4,8-PeCDF".

The concentration of each homologue and their total shall be calculated with the concentrations of the compound detected.

These expression methods shall be given in table 7 and the minimum limit of detection and the minimum limit of determination for sample gas shall be recorded clearly. When the results are to be summarized using the toxicity equivalency (TEQ) in 8.3, then attached table 6 and attached table 7 should be used as reference.

- b) **DL-PCBs** The concentrations of each compound and total concentration of them shall be recorded by the same procedures as described a) in the report of the analytical result of DL-PCBs. These expression methods shall be given in table 8 and the minimum limit of detection and the minimum limit of determination for sample gas shall be recorded clearly. When the results are to be summarized by using the toxicity equivalency (TEQ) in 8.3, then attached table 6 and attached table 7 should be used as reference.

8.2 Unit of concentration The found values of dioxins shall be expressed in ng/m³ (0 °C, 101.32 kPa).

8.3 Change to toxicity equivalency (TEQ) The concentration of dioxins are converted to toxicity equivalency (TEQ) by multiplying it with the toxicity equivalency factor (TEF) and expressing it in ng-TEQ/m³ (0 °C, 101.32 kPa).

- a) **The toxicity equivalency factor (TEF)** Unless otherwise specified, toxicity equivalency of dioxins shall be obtained from TEF in table 9 when other TEF is used, the TEF used shall be clearly mentioned in note, etc.
- b) **Calculation of toxicity equivalency (TEQ)** Calculate toxicity equivalency of each compound and take the total of them as the toxicity equivalency. Calculation should be as follows. The calculation procedures in case shall be recorded clearly without fail.
 - 1) Unless otherwise specified, values of not less than the minimum limit of determination shall be used as they are. Values which are between less than the minimum limit of determination and not less than the minimum limit of detection, and the values less than the minimum limit of detection shall be taken as zero, and the toxicity equivalency of each compound shall be calculated. The total toxicity equivalency shall be calculated by combining the toxicity equivalency of each compound.
 - 2) According to the purpose of the calculation for toxicity equivalency, the following methods may be used in addition to 1), and if specified, the following procedures shall be employed.
 - 2.1) The toxicity equivalency of each compound shall be calculated with an intact value which is not less than the minimum limit of determination, and an intact value between less than the minimum limit of determination and not less than the minimum limit of detection respectively. For values less than the minimum limit of detection, the toxicity equivalency of each compound is calculated with the value of the minimum limit of detection for the sample gas. The total toxicity equivalency shall be calculated by combining the toxicity equivalency of each compound.
 - 2.2) The toxicity equivalency of each compound shall be calculated with an intact value which is not less than the minimum limit of determination, and an intact value between less than the minimum limit of determination and not less than the minimum limit of detection respectively. For values less than the minimum limit of detection, the toxicity equivalency of each compound is calculated with the 1/2 value of the minimum limit of detection for the sample gas. The total toxicity equivalency shall be calculated by combining the toxicity equivalency of each compound.

8.4 Treatment of numerical values The treatment of the numerical values in the expression of the concentration shall be as follows, unless otherwise specified.

- a) The values of the concentration shall be rounded off according to **JIS Z 8401** and significant figures expressed as 2-digit figures. If the value is less than the minimum limit of detection, it shall be expressed that is less than the minimum limit of detection. However, the place to be marked are up to digit number of the minimum limit of detection for sample gas and the lower digit than the limit shall not be expressed.
- b) For the detection limit, the figures shall be rounded off in accordance with **JIS Z 8401** and expressed with significant figures of one digit number.

- c) For the calculation of toxicity equivalency, the TEQ of each compound is calculated from the measurement concentration marked, and the total value shall be rounded off in the same manner as in a) with 2-digit significant figures. In other words, rounding off is not applied to the toxicity equivalency of each compound.

Table 7 Expression method of PCDDs, PCDFs of homologues and compounds

Substituted compound by chlorine	PCDDs		PCDFs	
	Homologues	Compounds	Homologues	Compounds
Tetrachlorinated compound	TeCDDs	2,3,7,8- Others	TeCDFs	2,3,7,8- Others
Pentachlorinated compound	PeCDDs	1,2,3,7,8- Others	PeCDFs	1,2,3,7,8- 2,3,4,7,8- Others
Hexachlorinated compound	HxCDDs	1,2,3,4,7,8- 1,2,3,6,7,8- 1,2,3,7,8,9- Others	HxCDFs	1,2,3,4,7,8- 1,2,3,6,7,8- 1,2,3,7,8,9- 2,3,4,6,7,8- Others
Heptachlorinated compound	HpCDDs	1,2,3,4,6,7,8- Others	HpCDFs	1,2,3,4,6,7,8- 1,2,3,4,7,8,9- Others
Octachlorinated compound	OCDD	1,2,3,4,6,7,8,9-	OCDF	1,2,3,4,6,7,8,9-
Σ (Tetrachlorinated compound to octachlorinated compound)	PCDDs	—	PCDFs	—
	PCDDs + PCDFs			

Table 8 Expression method of the compounds of DL-PCBs

	Non-ortho compounds	Mono-ortho compounds
Tetrachlorinated compound (TeCB)	3,3',4,4'-(#77) 3,4,4',5-(#81)	
Pentachlorinated compound (PeCB)	3,3',4,4',5-(#126)	2,3,3',4,4'-(#105) 2,3,4,4',5-(#114) 2,3',4,4',5-(#118) 2',3,4,4',5-(#123)
Hexachlorinated compound (HxCB)	3,3',4,4',5,5'-(#169)	2,3,3',4,4',5'-(#156) 2,3,3',4,4',5-(#157) 2,3',4,4',5,5'-(#167)
Heptachlorinated compound (HpCB)		2,3,3',4,4',5,5'-(#189)
Σ (Tetrachlorinated compound to octachlorinated compound)	Total of non-ortho compounds	Total of mono-ortho compounds
	Total of DL-PCBs	

Table 9 Toxicity equivalency factor

Compounds		TEF (1998)*
PCDDs	2, 3, 7, 8-TeCDD	1
	1, 2, 3, 7, 8-PeCDD	1
	1, 2, 3, 4, 7, 8-HxCDD	0.1
	1, 2, 3, 6, 7, 8-HxCDD	0.1
	1, 2, 3, 7, 8, 9-HxCDD	0.1
	1, 2, 3, 4, 6, 7, 8-HpCDD	0.01
	1, 2, 3, 4, 6, 7, 8, 9-OCDD	0.000 1
	Others	0
PCDFs	2, 3, 7, 8-TeCDF	0.1
	1, 2, 3, 7, 8-PeCDF	0.05
	2, 3, 4, 7, 8-PeCDF	0.5
	1, 2, 3, 4, 7, 8-HxCDF	0.1
	1, 2, 3, 6, 7, 8-HxCDF	0.1
	1, 2, 3, 7, 8, 9-HxCDF	0.1
	2, 3, 4, 6, 7, 8-HxCDF	0.1
	1, 2, 3, 4, 6, 7, 8-HpCDF	0.01
	1, 2, 3, 4, 7, 8, 9-HpCDF	0.01
	1, 2, 3, 4, 6, 7, 8, 9-OCDF	0.000 1
	Others	0
DL-PCBs Non-ortho compounds	3, 4, 4', 5-TeCB (#81)	0.000 1
	3, 3', 4, 4'-TeCB (#77)	0.000 1
	3, 3', 4, 4', 5-PeCB (#126)	0.1
	3, 3', 4, 4', 5, 5'-HxCB (#169)	0.01
DL-PCBs Mono-ortho compounds	2', 3, 4, 4', 5-PeCB (#123)	0.000 1
	2, 3', 4, 4', 5-PeCB (#118)	0.000 1
	2, 3, 3', 4, 4'-PeCB (#105)	0.000 1
	2, 3, 4, 4', 5-PeCB (#114)	0.000 5
	2, 3', 4, 4', 5, 5'-HxCB (#167)	0.000 01
	2, 3, 3', 4, 4', 5-HxCB (#156)	0.000 5
	2, 3, 3', 4, 4', 5'-HxCB (#157)	0.000 5
	2, 3, 3', 4, 4', 5, 5'-HpCB (#189)	0.000 1

Note * TEF refers to the proposal of WHO/IPCS in 1998.

9 Quality control of the measured data As dioxins are measured in extremely low concentrations, it is necessary to control the precision of analysis sufficiently. Quality control of the measured data shall be as follows.

9.1 Assurance of reliability of measured data

9.1.1 Confirmation for recovery ratio of internal standard substance

- a) **Recovery ratio of internal standard substance for sampling spike** The recovery ratio of the internal standard substance for sampling spike shall be checked. When the recovery ratio of each sampling spike is out of the range between 70 % and 130 %, find the causes, change them for improvements and carry out sampling of the sample gas again.

- b) **Recovery ratio of internal standard substance for the cleanup spike** The recovery ratio of the internal standard substance for cleanup spike shall be checked. When the recovery ratio of each cleanup spike is out of the range between 50 % and 120 %, the cleanup procedures from extracted solution shall be carried out again.

9.1.2 Confirmation of the minimum limit of detection and the minimum limit of determination

- a) **The minimum limit of detection and the minimum limit of determination for the apparatus** Measure the standard solution of the minimum concentration among the solutions for the calibration curve (containing each standard substance such as 0.1 pg to 0.5 pg for tetra- and penta-chlorinated compounds, 0.2 pg to 1.0 pg for hexa- and hepta-chlorinated compounds, 0.5 pg to 2.5 pg for octa-chlorinated compounds and 0.2 pg to 1.0 pg of DL-PCBs) using the GC/MS, and determine each isomer substituted by chlorine at 2,3,7,8-positions. Repeat this operation 5 times or more and calculate the standard deviation by using the measured values obtained. Take threefold of the standard deviation as the minimum limit of detection for the apparatus and tenfold of the standard deviation as the minimum limit for the determination.

When the minimum limits of detection for the apparatus are larger than 0.1 pg for tetra- and penta-chlorinated compounds, 0.2 pg for hexa- and hepta-chlorinated compounds, 0.5 pg for octa-chlorinated compounds and 0.2 pg for DL-PCBs, check the equipment and the apparatus, and adjust them to give values to be equal or smaller than these values.

As the minimum limit of detection and the minimum limit of the determination for the apparatus are liable to vary depending on the conditions of GC/MS used, check them at regular intervals and control the apparatus so that it always gives satisfactory results. It shall definitely be checked when the GC/MS used and conditions of measurement have been changed.

- b) **The minimum limit of detection and the minimum limit of determination for the measuring method** Add standard substance of such volume that the injected volume to GC/MS is equal to that of the minimum limit of determination for the apparatus, into the extracted solution obtained from the same volume of the absorption solution, extracted solution from adsorbent and filter used in the measurement. Then carry out pretreatment, measurement, identification and determination of the sample by GC/MS. Repeat these operations 5 times or more, calculate the standard deviation with data obtained. Take threefold of the standard deviation as the minimum limit of detection for the measuring method and tenfold of the standard deviation as the minimum limit of the determination for the measuring method.

Because the minimum limit of detection and the minimum limit of determination for the measuring method are liable to vary depending on the conditions of pretreatment and measurement used, check them at regular intervals and control them so that it always gives the satisfactory results. It shall definitely be confirmed when pretreatment operations and conditions of measurement have been changed.

- c) **The minimum limit of detection and the minimum limit of determination for the sample gas** Because those limits of the sample gas are liable to vary depending on the sampling volume, etc., they shall be obtained for each sample by using the minimum limit of detection and the minimum limit of determination of measurement method. The minimum limit of detection of the sample gas obtained shall be not more than 1/30 of the concentration to be evaluated.
- d) **Confirmation of the minimum limit of detection at the sample measurement** When measuring the actual sample, if the peak can not be detected for the isomers substituted by chlorine at 2,3,7,8-positions and DL-PCBs, estimate the minimum limit of detection at the sample measurement with the noise width of the base line close to the peak on its chromatogram. Each numerical value calculated shall be equal or smaller than those of the minimum limit of detection for the measuring method.

When the numerical value is beyond that of the minimum limit of detection, check for existence of the problems in pretreatment and measuring operations, eliminate the cause, and repeat the measurement again. The concentration in sample gas calculated from the minimum limit of detection at the time of measurement of sample shall be made so as to be 1/30 of the concentration to be evaluated at least.

9.1.3 Blank test There are 2 types of the blank test. One of them shall be carried out to check the level of contamination such as filters, adsorbent, absorption solution and the reagents used at the pretreatment operation (hereafter referred to as "operational blank test"). The other shall be carried out to check the contamination in sampling of the sample gas and carrying the sample (hereafter referred to as "travel blank test").

Remarks : If the blank values are high, these not only decrease the sensitivity of the measurement, but also lower the reliability of the measured values. For this reason, blank test value shall be kept to a minimum. Therefore, where necessary, the pretreatment should preferably be carried out in the clean draft.

- a) **Operational blank test** The operational blank test shall be carried out for the purpose of checking for contamination, which is caused during preparation of the sample or injection into GC/MS, and for setting up an environment free from interferences for the measurement. Prepare a filter, adsorbent, absorption solution of the same lot as those for sampling, and carry out the same procedures in clauses 6 and 7 as those for the sample.

This test shall be periodically carried out in a definite period when the lot of reagents, etc. is changed, and be sufficiently controlled against the contamination when operated. Furthermore, that is carried out prior to measurement in the following cases and the result of operation blank test should be sufficiently made low.

- 1) In the case of a significant alteration in the pretreatment is to be made such as to use new reagents or apparatus, or to reuse the repaired apparatus.
- 2) In the case where a sample of high concentration that is likely to be contaminated is to be measured.

- b) **Travel blank test** The travel blank test shall be carried out to check the existence of contamination from the preparation of the sampling for the sample gas to the transportation of sample taken. Carry out the solution, which is handled and transported, with the same procedures as the sample analysis in clauses 6 and 7 without those of the sampling.

If there (such as ash gathered in an electric precipitator) is a possibility of contamination from sampling to the transportation of sample taken, the travel blank test shall be carried out. Otherwise, if it is confirmed that the prevention of contamination is surely carried out, it is not necessary to carry out travel blank test every time. However, in order to make sure of the reliability in the sampling, sufficient consideration shall be given to these values of the travel blank test and the data shall be secured for future presentation if required.

Travel blank tests are carried out at least three samples shall be tested. The average value (e) shall be calculated to correct the measurement values as below.

- 1) If the average of the travel blank values (e) (hereafter referred to as "travel blank value") are considered to be the same as the operation blank value (a) (equal or less) ($e \leq a$), any contamination during travel can be disregarded.
- 2) If the travel blank value (e) is more than the operation blank value (a) ($e > a$), calculation shall be as follows.
 - 2.1) If the travel blank value (e) is less than the measurement value of the sample (d), ($d \geq e$), and the measurement value (d) is more than the concentration value (f) which is calculated with ten times value of the standard deviation(s) obtained at travel blank test ($d \geq f$), the concentration shall be calculated by subtracting the travel blank value (e) from the measurement value (d).
 - 2.2) If the measurement value (d) is less than the concentration value (f), which is calculated with ten times value of the standard deviation(s) obtained at travel blank test ($d < f$), or the travel blank value (e) is more than the measurement value (d) of the sample ($e > d$), there will be a problem in the reliability for the measured values, in principle, these are considered as missing measurements. In this case, look for the cause of the contamination, eliminate it, and repeat sampling.

9.1.4 Dual measurement For the dual measurement, the same gas shall be collected simultaneously with 2 apparatuses. This sampling shall be carried out at a frequency of approximately 1 out of 10 samples if possible. The average for the measured value of the compounds detected by not less than the minimum limit of determination is calculated with isomers substituted by chlorine at 2,3,7,8-positions and the DL-PCBs. It shall be confirmed that each measured value is within $\pm 30\%$ of the average values.

If it is confirmed that the operation of sampling of sample gas is sufficiently examined and the reliability is sufficiently ensured, the sampling of sample gas for double measurement needs not to carry out at the frequency mentioned above, however, the reliability in the sampling and the data shall be secured for future presentation if required.

9.1.5 Standard substance For the ensuring of the reliability of measured values, the measurement hardware shall be periodically calibrated by using the standard substance which is traceable to the national measurement standard or recognized by the national organization for measurement standard. These standard solutions shall be packed in the glass sealed containers for preventing a change in the concentration due to the solvent vapourizing, and shall be stored in a cool place and shielded from light under controlled conditions.

9.2 Requirements for operation of the measurement

9.2.1 Gas sampling The following items shall be taken into consideration when sampling.

- a) **Preparation and storage of sampling apparatus** Cylindrical filter paper, resin, and absorption solution shall be sufficiently washed prior to use and the apparatuses without the blank test values shall be employed. In particular, adsorbents shall be stored in an airtight container to prevent outside contamination during the procedure from washing for preparation to gas sampling.
- b) **Gas sampling apparatus** Tool and parts used in each piece of sampling apparatus shall be washed to reduce contamination caused by tool and other instruments. No leakage of the apparatus shall be confirmed by the operations that all components of the sampling apparatus are fixed and checked the airtightness. The part collecting dioxins shall be shaded.
- c) **Gas meter** For ensuring of the reliability of gas meter, the measurement hardware shall be periodically calibrated by using the measurement standard which is traceable to the national measurement standard or recognized by the national organization for measurement standard.
- d) **Sampling of the representative sample** It is important that the sampling of representative sample shall be appropriate to the purpose for sampling of the sample gas. In principle, sampling shall be begun at least 1 h after the combustion conditions becomes stable and the measurement of exhaust gas, in the incinerator with continuous operation, shall be carried out among an average of 4 h.

In a furnace with intermittent operation, if the exhaust gas in the steady state flow is taken as the representative sample, sampling shall be done during normal operation excluding start-up and shutdown. If the sampling is given the serious effects of start-up and shutdown, it shall be carried out including them or in accordance with the optional conditions.

In the gas sampling, temperature, concentration of carbon dioxide, etc. should be measured continuously, recorded the variations from starting to finishing of sampling and attached these data to a report.

- e) **Sampling procedures of the sample gas** In the sampling of the sample gas, it is important that there is no secondary formation or loss of dioxins in the collecting part during sampling and that a sufficient amount of dioxins will be recovered from the samples after sampling. The sampling of the sample gas shall be carried out with the isokinetic sampling in accordance with **JIS Z 8808**. Flow rate, temperature, pressure, moisture volume, components and other factors

shall be measured prior to the sampling of the sample gas. Isokinetic sampling flow rate shall be calculated and a nozzle of appropriate size (inner diameter of 4 mm or more) shall be attached.

- f) **Storage and transportation of the samples** The samples collected shall be stored in an air-tight container to prevent contamination by the surrounding atmosphere and leakage. Sample should also be shielded from light when storing and transporting.

9.2.2 Pretreatment Attention shall be paid to the following items in the pretreatment operation.

- a) **Extraction of the sample**
 - 1) **Liquid-liquid extraction** Choose the solvent and check the extraction conditions carefully so that extraction can take place to the target solvent layer.
 - 2) **Soxhlet extractor** In the Soxhlet extraction, the filter paper made from glass fibre shall be dried sufficiently.
 - 3) For preventing the decomposition by light, it shall be avoided that strong light shines upon the sample. When light shines especially by Soxhlet extraction for a long time, it shall be carried out as shading.
- b) **Operation of the sulfuric acid treatment—Silica gel column chromatograph or multilayer silica gel column chromatograph**
 - 1) Check that the extracted solution is not be coloured completely.
 - 2) Since the fraction conditions of a chromatography varies depending on the type and activity of the filling material, or the type and quantity of the solvent, carry out fraction tests using such a fly as extraction solution, which is included all type compounds, to determine the optimum conditions previously.
- c) **Operation of the alumina column chromatograph** Alumina polarity can vary significantly depending on the manufacturing lot, term and conditions of storage after opening the seal. By using alumina with lowered activity for dioxins, 1,3,6,8-TeCDD and 1,3,6,8-TeCDF may be eluted in the first fraction, octachlorinated compounds may not be eluted with specified volume of hexane solutions with 50 vol% dichloromethane, and some parts of the DL-PCBs are eluted into the eluted fraction by hexane. Therefore, the fraction tests shall be carried out using such a solution as obtained by the extraction of fly ash, which includes all type compounds, in order to determine the optimum conditions.

9.2.3 Identification and determination Attention shall be paid to the following items on identification and determination.

- a) **Gas chromatograph mass spectrometer** The measurement conditions of a gas chromatograph mass spectrometer shall be set according to each purpose and the apparatus shall be adjusted so as to be capable of determining the samples. It shall be checked so that a full reliable analysis can be carried out by investigating response linearity, stability, existence and scale of interference, which may cause measurement errors, and methods of correcting them.

- 1) **Adjustment of gas chromatograph** The column tank temperature, column injection temperature, flow rate of carrier gas, and other conditions shall be set and confirmed to ensure that the response is stable, the retention time of each chlorinated compound is within the appropriate range and the peaks are separate sufficiently. The splitless time, flow rate of purge gas and other values shall also be set.

If the target components for analysis are not sufficiently separated from the other components, replace the capillary column with a new one. If the problem disappears when the capillary column is cut by approximately 300 mm (at one end or both ends), the column is not necessarily replaced.

- 2) **Adjustment of the mass spectrometer** Introduce the standard substance for mass calibration (perfluorokerosene: PFK) to the mass spectrometer, calibrate mass pattern and resolution (10 000 or more) with a mass calibration program. In addition, conduct a check on the basic points such as the sensitivity to the apparatus, etc. Record and store the results of the tuning.

- 3) **Operation condition of the gas chromatograph mass spectrometer** While the width of the peak obtained with using a capillary column is approximately 5 s to 10 s, in order to obtain enough data points to make up the peak, even if the peak of single component on chromatogram is the peak of the narrowest width, the sampling period of selected ion detection shall be set up so that the measurement point constituting the peak is made not less than 7 points. The number of monitor channels that can be set in one analysis depends on the sensitivity required and shall therefore be set with the utmost care.

It is acceptable to carry out grouping by time shearing if consideration is given to the retention time of each peak on the chromatogram. However, in this case appropriate conditions shall be set so that a relevant internal standard substance peak appears for each group.

- 4) **Maintenance and management of the apparatus** For maintaining the performance of the gas chromatograph mass spectrometer, daily check shall be carried out regularly. In particular, as the contamination at the interface of the gas chromatograph and in the ionization chamber causes a decrease in sensitivity, the resolution and the precision of the measurement, cleaning shall be carried out appropriately.
- b) **Preparation of calibration curve** The calibration curve shall be prepared when the measurement is started at first time and afterwards that shall be prepared again in the case of changing on measurement such as when the standard solution is renewed, when the analytical conditions are altered, and so on or in the case of large variation of sensitivity.

For the maintaining of the precision of measurement, it is recommended to renew periodically at the time other than that mentioned above. The degree of period for renewal is different depending on the condition of measurement, the operation condition of apparatus, etc., therefore, that shall be determined beforehand according to a definite period such as three month or a definite number of sample for measurement from the condition of sensitivity variation, etc.

- c) **Fluctuations in the sensitivity of the apparatus** Measure periodically, at least once a day, one or more concentration of standard solution and check that there are no remarkable changes in the fluctuation for the sensitivity of internal standard substance in comparison with that obtained at the time when the calibration curve was prepared.

Check that the fluctuation of the relative response factor for each isomers of dioxins to the internal standard substance is within $\pm 10\%$ compared to the relative response factor at the time the calibration curve is prepared. If fluctuations exceed this range, eliminate the cause for this, prepare the calibration curve again and repeat the measurement of samples.

For fluctuations in the retention time, in the case of gradual fluctuations caused by the deterioration of the separation column, these can be dealt with a required. If the fluctuations occur within a relatively short period (normally fluctuation in the retention time is more than $\pm 5\%$ in a day, or the relative retention time ratio to that of the internal standard substance is more than $\pm 2\%$), eliminate the cause for these and repeat the measurement of samples used before the adjustment.

9.2.4 Handling of abnormal values and missing values If the sensitivity fluctuation of the analytical apparatus is high, the travel blank values are high, and the dual measurement results vary widely, there must be a problem for the reliability of the measured values and it is necessary to repeat the measurement and sampling of samples considered to be erroneously measured. These problems lead not only to the consumption of manpower, time, and costs, but also to a negative effect on the evaluation of the overall measurement results. For this reason, a full preliminary survey must be carried out to ensure no abnormal values nor missing values. Moreover, it is extremely important to fully investigate the details of the occurrence of abnormal values or missing value, and to keep a record of these circumstances in order to prevent similar problems from occurring in the future.

9.3 Record of measurement operation The following information shall be recorded, arranged and stored.

- a) Adjustment, calibration, and operation of equipment and tools used for sampling of sampling gas.
- b) Preparation, handling, and storage conditions of containers, resin, absorption solution, collection filters, etc.
- c) Conditions during sampling of sample gas.

Sampling method, sampling points, sampling date and time, temperature, moisture content, static pressure, flow speed, wet and dry flow rate, the results of the leakage test, and colouring of sampling system.
- d) Gas sampling conditions.

Suction volume, sampling time, sampling quantity, etc.
- e) Calibration and operation of measuring apparatus.
- f) Record the operation from pretreatment to measurement
- g) All available information related to the data acquisition of measurement value.

9.4 Report on the precision control The following information related to the precision control shall be recorded and submit the report with data, if necessary.

- a) Record of the traceability and calibration of gas meter.
- b) Record of daily check and adjustments for a gas chromatograph mass spectrometer (calibration of the apparatus, etc.).
- c) Setting of the measurement conditions and its result for the measuring equipment.
- d) Manufacturers and traceability of standard substances, etc.
- e) Measurement results for minimum limits of detection and determination.
- f) Results of operational blank test and travel blank test.
- g) Verification results of recovery tests such as sampling and pretreatment, etc.
- h) Fluctuations of the sensitivity in measurement apparatus.
- i) Record of measurement operations (records of sampling to pretreatment, and measurement).

Attached Table 1 An example of measuring conditions in the gas chromatograph mass spectrometer for PCDDs and PCDFs

Gas chromatograph	① Target compounds for determination Homologue of TeCDDs, TeCDFs, PeCDFs and isomers substituted by chlorine at 2,3,7,8-positions Column type: SP-2331 (made by Supelco) Inside diameter 0.25 mm, length 60 m, thickness 0.20 μm Column temperature: 120 °C (1 min) $\rightarrow$ (50 °C/min) $\rightarrow$ 200 °C $\rightarrow$ (2 °C/min) $\rightarrow$ 260 °C (25 min) Column injection port temperature: 260 °C Sample injection method: Splitless (90 s) Sample injection volume: 1 μl ② Target compounds for determination Homologue of PeCDDs, HxCDDs, HxCDFs and isomers substituted by chlorine at 2,3,7,8-positions Column type: SP-2331 (made by Supelco) Inside diameter 0.25 mm, length 60 m, thickness 0.20 μm Column temperature: 120 °C (1 min) $\rightarrow$ (50 °C/min) $\rightarrow$ 200 °C $\rightarrow$ (2 °C/min) $\rightarrow$ 260 °C (25 min) Column injection port temperature: 260 °C Sample injection method: Splitless (90 s) Sample injection volume: 1 μl ③ Target compounds for determination Homologue of HpCDDs, OCDD, HpCDFs, OCDF and isomers substituted by chlorine at 2,3,7,8-positions Column type: DB-17 (made by J&W) Inside diameter 0.32 mm, length 30 m, thickness 0.25 μm Column temperature: 120 °C (1 min) $\rightarrow$ (20 °C/min) $\rightarrow$ 160 °C $\rightarrow$ (3 °C/min) $\rightarrow$ 280 °C (5 min) Column injection port temperature: 280 °C Sample injection method: Splitless (90 s) Sample injection volume: 1 μl
Mass spectrometer	Resolution: 10 000 or more Electron accelerating voltage: 70 V Ionization current: 1 mA Ion source temperature: 260 °C Detection method: SIM method with Lockmass mode

Attached Table 2 An example of measuring conditions in the gas chromatograph mass spectrometer for PCDDs and PCDFs

Gas chromatograph	① Target compounds for determination Homologue of TeCDDs to HxCDDs, TeCDFs to HxCDFs and isomers substituted by chlorine at 2,3,7,8-positions Column type: SP-2331 (made by Supelco) Inside diameter 0.25 mm, length 60 m, thickness 0.20 μm Column temperature: 120 °C (1 min) $\rightarrow$ (50 °C/min) $\rightarrow$ 200 °C $\rightarrow$ (2 °C/min) $\rightarrow$ 260 °C Column injection port temperature: 260 °C Sample injection method: Splitless (60 s) Sample injection volume: 1 μl ② Target compounds for determination Homologue of HpCDDs, OCDD, HpCDFs, OCDF and isomers substituted by chlorine at 2,3,7,8-positions Column type: DB-5 (made by J&W) Inside diameter 0.32 mm, length 30 m, thickness 0.25 μm Column temperature: 120 °C (1 min) $\rightarrow$ (50 °C/min) $\rightarrow$ 180 °C $\rightarrow$ (3 °C/min) $\rightarrow$ 280 °C Column injection port temperature: 300 °C Sample injection method: Splitless (60 s) Sample injection volume: 1 μl
Mass spectrometer	Resolution: 10 000 or more Electron accelerating voltage: 70 V Ionization current: 1 mA Ion source temperature: 270 °C Detection method: SIM method with Lockmass mode

Attached Table 3 An example of measuring conditions in the gas chromatograph mass spectrometer for PCDDs and PCDFs

Gas chromatograph	① Target compounds for determination Total of each homologue of TeCDDs to OCDDs, TeCDFs to OCDFs and isomers substituted by chlorine at 2,3,7,8-positions Column type: SP-2331 (made by Supelco) Inside diameter 0.25 mm, length 60 m, thickness 0.20 μm Column temperature: 120 °C (1 min) $\rightarrow$ (50 °C/min) $\rightarrow$ 200 °C $\rightarrow$ (2 °C/min) $\rightarrow$ 260 °C (30 min) Column injection port temperature: 170 °C $\rightarrow$ 300 °C (100 °C/min) Sample injection method: On column injection method Sample injection volume: 1 μl	
	② Target compounds for determination Total of each homologue of TeCDDs to OCDDs, TeCDFs to OCDFs and isomers substituted by chlorine at 2,3,7,8-positions Column type: DB-17 (made by J&W) Inside diameter 0.32 mm, length 30 m, thickness 0.25 μm Column temperature: 120 °C (1 min) $\rightarrow$ (20 °C/min) $\rightarrow$ 160 °C $\rightarrow$ (3 °C/min) $\rightarrow$ 280 °C Column injection port temperature: 150 °C $\rightarrow$ 300 °C (100 °C/min) Sample injection method: On column injection method Sample injection volume: 1 μl	
Mass spectrometer	Resolution:	10 000 or more
	Electron accelerating voltage:	35 V to 40 V
	Ionization current:	0.5 mA
	Ion source temperature:	270 °C
	Detection method:	SIM method with Lockmass mode

Attached Table 4 An example of measuring conditions in the gas chromatograph mass spectrometer for DL-PCBs

Gas chromatograph	Target compounds for determination DL-PCBs Column type: DB-5 ms (made by J&W) Inside diameter 0.32 mm, length 60 m, thickness 0.25 μm Column temperature: 150 °C (1 min) $\rightarrow$ (20 °C/min) $\rightarrow$ 185 °C (3 min) (2 °C/min) $\rightarrow$ 245 °C (3 min) (6 °C/min) $\rightarrow$ 290 °C (10 min) Column injection port temperature: 170 °C $\rightarrow$ 300 °C (100 °C/min) Sample injection method: On column injection method
Mass spectrometer	Resolution: 10 000 or more Electron accelerating voltage: 35 V to 40 V Ionization current: 0.5 mA Ion source temperature: 270 °C Detection method: SIM method with Lockmass mode

Attached Table 5 An example of measuring conditions in the gas chromatograph mass spectrometer for DL-PCBs

Gas chromatograph	Target compounds for determination DL-PCBs Column type: HT8 (made by SGE) Inside diameter 0.22 mm, length 50 m, thickness 0.25 μm Column temperature: 130 °C (1 min) $\rightarrow$ (20 °C/min) $\rightarrow$ 220 °C (5 °C/min) $\rightarrow$ 320 °C Column injection port temperature: 280 °C Sample injection method: Splitless method (90 s)
Mass spectrometer	Resolution: 10 000 or more Electron accelerating voltage: 30 V to 50 V Ionization current: 0.5 mA Ion source temperature: 335 °C Detection method: SIM method with Lockmass mode

Attached Table 6-1 Sample format for the results of the measurement

I Results of the measurement for PCDDs and PCDFs in exhaust gas

		The final discharge exit (0 °C, 101.32 kPa)				
		Concentration of actual measurement	Minimum limit of determination in sample gas	Minimum limit of detection in sample gas	Toxicity equivalency factor	Toxicity equivalency (TEQ)
		ng/m ³	ng/m ³	ng/m ³		ng-TEQ/m ³
PCDDs	2,3,7,8-TeCDD				1	
	TeCDDs				—	
	1,2,3,7,8-PeCDD				1	
	PeCDDs				—	
	1,2,3,4,7,8-HxCDD				0.1	
	1,2,3,6,7,8-HxCDD				0.1	
	1,2,3,7,8,9-HxCDD				0.1	
	HxCDDs				—	
	1,2,3,4,6,7,8-HpCDD				0.01	
	HpCDDs				—	
	OCDD				0.000 1	
	Total PCDDs		—	—	—	
PCDFs	2,3,7,8-TeCDF				0.1	
	TeCDFs				—	
	1,2,3,7,8-PeCDF				0.05	
	2,3,4,7,8-PeCDF				0.5	
	PeCDFs				—	
	1,2,3,4,7,8-HxCDF				0.1	
	1,2,3,6,7,8-HxCDF				0.1	
	1,2,3,7,8,9-HxCDF				0.1	
	2,3,4,6,7,8-HxCDF				0.1	
	HxCDFs				—	
	1,2,3,4,6,7,8-HpCDF				0.01	
	1,2,3,4,7,8,9-HpCDF				0.01	
	HpCDFs				—	
	OCDF				0.000 1	
	Total PCDFs		—	—	—	
Total (PCDDs+PCDFs)			—	—	—	

- Remarks 1 In the column of concentration of actual measurement, the figures in the parentheses show the concentration of the minimum limit of detection or over to and excluding the minimum limit of determination.
- 2 ND in the actual concentration shows less than the minimum limit of detection.
- 3 Toxicity equivalency factors are applied TEF of WHO/IPCS (1998).
- 4 Toxicity equivalency is calculated as zero for the actual concentration of less than the minimum limit of determination.

Attached Table 6-2 Sample format for the results of the measurement

II Results of the measurement for DL-PCBs in exhaust gas

		The final discharge exit (0 °C, 101.32 kPa)				
		Concentration of actual measurement	Minimum limit of determination in sample gas	Minimum limit of detection in sample gas	Toxicity equivalency factor	Toxicity equivalency (TEQ)
		ng/m ³	ng/m ³	ng/m ³		ng-TEQ/m ³
Non-ortho compounds	3,4,4',5-TeCB (# 81)				0.000 1	
	3,3',4,4'-TeCB (# 77)				0.000 1	
	3,3',4,4',5-PeCB (#126)				0.1	
	3,3',4,4',5,5'-HxCB (#169)				0.01	
	Total				—	
Mono-ortho compounds	2',3,4,4',5-PeCB (#123)				0.000 1	
	2,3',4,4',5-PeCB (#118)				0.000 1	
	2,3,3',4,4'-PeCB (#105)				0.000 1	
	2,3,4,4',5-PeCB (#114)				0.000 5	
	2,3',4,4',5,5'-HxCB (#167)				0.000 01	
	2,3,3',4,4',5-HxCB (#156)				0.000 5	
	2,3,3',4,4',5'-HxCB (#157)				0.000 5	
	2,3,3',4,4',5,5'-HpCB (#189)				0.000 1	
Total					—	
Total DL-PCBs			—	—	—	

- Remarks
- 1 In the column of concentration of actual measurement, the figures in the parentheses show the concentration of the minimum limit of detection or over to and excluding the minimum limit of determination.
 - 2 ND in the actual concentration shows less than the minimum limit of detection.
 - 3 Toxicity equivalency factors are applied TEF of WHO/IPCS (1998).
 - 4 Toxicity equivalency is calculated as zero for the actual concentration of less than the minimum limit of determination.

Attached Table 7-1 Sample format for the results of the measurement

I Results of the measurement regarding PCDD and PCDF substituted by chlorine at 2,3,7,8-positions and DL-PCBs in exhaust gas

		The final discharge exit (0 °C, 101.32 kPa)			
		Concentration of actual measurement	Converted concentration to 12 % oxygen	Toxicity equivalency factor TEF	Toxicity equivalency (TEQ)
		Cs (ng/m ³)	C (ng/m ³)		(ng-TEQ/m ³)
PCDD	2,3,7,8-TeCDD			1	
	1,2,3,7,8-PeCDD			1	
	1,2,3,4,7,8-HxCDD			0.1	
	1,2,3,6,7,8-HxCDD			0.1	
	1,2,3,7,8,9-HxCDD			0.1	
	1,2,3,4,6,7,8-HpCDD			0.01	
	OCDD			0.000 1	
	Total PCDDs	—	—	—	
PCDF	2,3,7,8-TeCDF			0.1	
	1,2,3,7,8-PeCDF			0.05	
	2,3,4,7,8-PeCDF			0.5	
	1,2,3,4,7,8-HxCDF			0.1	
	1,2,3,6,7,8-HxCDF			0.1	
	1,2,3,7,8,9-HxCDF			0.1	
	2,3,4,6,7,8-HxCDF			0.1	
	1,2,3,4,6,7,8-HpCDF			0.01	
	1,2,3,4,7,8,9-HpCDF			0.01	
	OCDF			0.000 1	
	Total PCDFs	—	—	—	
Total (PCDDs + PCDFs)		—	—	—	
DL-PCBs	3,4,4',5-TeCB (# 81)			0.000 1	
	3,3',4,4'-TeCB (# 77)			0.000 1	
	3,3',4,4',5-PeCB (#126)			0.1	
	3,3',4,4',5,5'-HxCB (#169)			0.01	
	Total non-ortho compounds				
	2',3,4,4',5-PeCB (#123)			0.000 1	
	2,3',4,4',5-PeCB (#118)			0.000 1	
	2,3,3',4,4'-PeCB (#105)			0.000 1	
	2,3,4,4',5-PeCB (#114)			0.000 5	
	2,3',4,4',5,5'-HxCB (#167)			0.000 01	
	2,3,3',4,4',5-HxCB (#156)			0.000 5	
	2,3,3',4,4',5'-HxCB (#157)			0.000 5	
	2,3,3',4,4',5,5'-HpCB (#189)			0.000 1	
	Total mono-ortho compounds			—	
Total DL-PCBs				—	
Total dioxins		—	—	—	

Attached Table 7-2 Sample format for the results of the measurement**II Results of the measurement for PCDDs and PCDFs homologue in exhaust gas**

		The final discharge exit (0 °C, 101.32 kPa)	
		Concentration of actual measurement Cs (ng/m ³)	Converted concentration to 12 % oxygen C (ng/m ³)
PCDD	TeCDDs		
	PeCDDs		
	HxCDDs		
	HpCDDs		
	OCDD		
	Total PCDDs		
PCDF	TeCDFs		
	PeCDFs		
	HxCDFs		
	HpCDFs		
	OCDF		
	Total PCDFs		
Total (PCDDs + PCDFs)			

Remarks 1 In the column of concentration of actual measurement, the figures in the parentheses show the concentration of the minimum limit of detection or over to and excluding the minimum limit of determination.

2 "< a" in the column of actual concentration shows less than the minimum limit of detection (α).

3 Converted concentrations to 12 % oxygen (C) are calculated with the following formula.

$$C = \frac{21 - 12}{21 - O_s} \times C_s \quad (O_s = \quad \%)$$

4 Toxicity equivalency factors are applied TEF of WHO/IPCS (1998).

5 Toxicity equivalency quantities are calculated as zero for the actual concentration of less than the minimum limit of determination.

6 The minimum limit of detection and the minimum limit of determination (0 °C, 101.32 kPa) in the sample gas of this measurement shall be as follows.

	Minimum limit of detection in sample gas (ng/m ³)	Minimum limit of determination in sample gas (ng/m ³)
TeCDD		
PeCDD		
HxCDD		
HpCDD		
OCDD		
TeCDF		
PeCDF		
HxCDF		
HpCDF		
OCDF		
DL-PCB		

Annex 1 (normative)

Sample gas sampling apparatus

1 Scope This annex specifies the sample gas sampling apparatus.

2 Classification and outline of sample gas sampling apparatus The classification and outline of sample gas sampling apparatus shall be as follows. The sample gas sampling apparatus as shown herein is the one having the actual results already used much or the one in which it is confirmed that the conditions specified in 5.2 of the text are satisfied. When even other apparatus than that shown herein is confirmed to satisfy the condition specified in 5.2 of the text, it can be used.

- a) **JIS type I apparatus** That is the apparatus to be a basis for the sampling apparatus in which the dioxins in exhaust gas are collected by means of the filtration-collection by filter, the liquid-collection by absorption bottle and the adsorption-filtration by adsorption column.
- b) **JIS type II apparatus** That is the apparatus in which the exhaust gas is collected by filtration and then, collected by adsorption by using cylindrical filter type adsorbent processed specially and moulded by alumina adsorbent.
- c) **JIS type III apparatus** That is the apparatus specified in European Standard (EN 1948), in which the exhaust gas is liquid-collected by cooling with cooling probe and then filtration-collection is carried out by means of adsorption-collection and filter.

3 JIS type I apparatus

3.1 Scope in use This apparatus can be applicable to all exhaust gas.

3.2 Construction of apparatus This apparatus shall be composed of sampling pipe part followed by filter-collecting part, liquid-collecting part (I), adsorption-collecting part and liquid-collecting part (II). The outline of this apparatus is as shown in annex 1 figure 1.

- a) **Filter-collecting part** For the filter-collecting part, the type II dust collecting device specified in 8.3 of JIS Z 8808 shall be used. In the case of using filter paper, the circular or cylindrical one made of silica fibre shall be used. In the case of dust tube, it shall be packed with glass fibre or silica fibre. In any case, it shall be confirmed that blank test component and other interfering component are not found in presence prior to the test.

When the dust content is little and the sampling and measurement are not hindered, the filter-collecting part may be omitted. And, depending on the kinds of combustion equipment, the dust tube containing silica fibre, etc. shall be used in front of cylindrical filter paper.

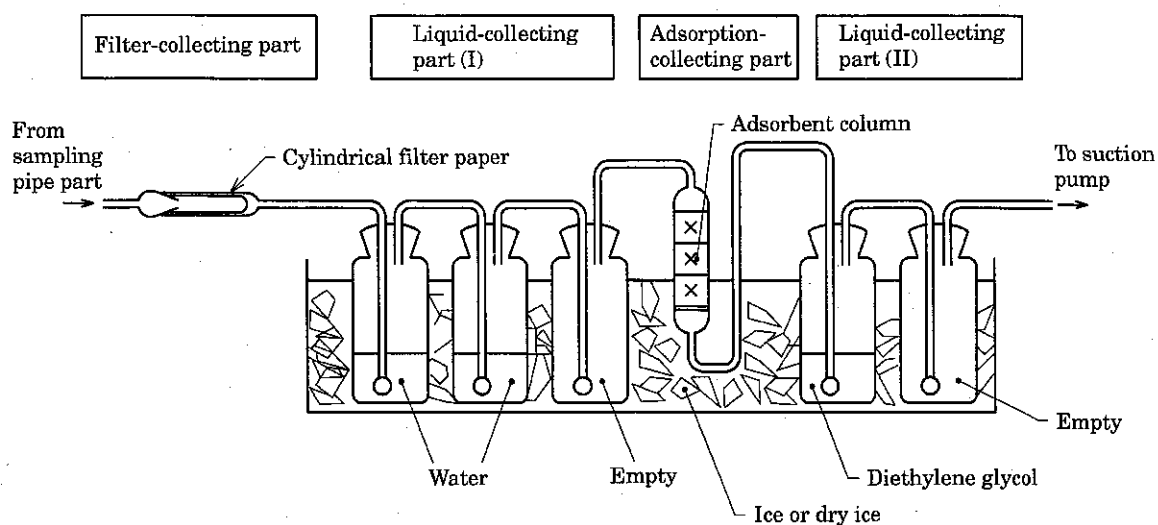
- b) **Liquid-collecting part** The absorbing bottles of 0.5 L to 1 L in inside capacity shall be connected in series and the two types such as that containing 100 ml

to 300 ml of hexane washings and that containing 100 ml to 300 ml of 2,2'-6 oxybisethanol (diethylene glycol) shall be used. For the collecting-part (I), hexane washings shall be put in the first piece and the second piece of absorbing bottles and the third piece of absorbing bottle be made empty, and for the collecting-part (II), diethylene glycol shall be put in the first piece and the second piece be made empty.

- c) **Adsorption-collecting part** The washed and dried adsorbent of approximately 40 g to 70 g shall be packed in a glass pipe (30 mm to 50 mm in inside diameter, 70 mm to 200 mm in length, 100 ml to 150 ml in capacity), which is made a column so that the both ends are fixed with glass wool, etc. so that the adsorbent does not move, and fixed by connecting vertically between the liquid-collecting part (I) and the liquid-collecting part (II).

If necessary, the adsorbent column may be used in two stages.

- d) **Connecting-part** The connecting pipe from filter-collecting part to liquid-collecting part shall be made as short as possible and that of glass or fluorocarbon resin shall be used. For the connection for each part, an interchangeable ball and socket joint pipe or fluorocarbon resin pipe joint, etc. shall be used and grease shall not be used at the connecting-part.



Annex 1 Figure 1 Outline of JIS type I sampling apparatus

3.3 Reagents The reagents to be used for this apparatus shall be as follows. It shall be confirmed by blank test, etc. that these reagents do not hinder the measurement.

- Water** Water of grade A4 (or A3) specified in JIS K 0557.
- Hexane** As specified in JIS K 8825 or that of equivalent quality.
- 2,2'-Oxybisethanol (diethylene glycol)** That of quality without the hindrance to measurement.
- Hexane washings** The water in a) washed sufficiently by hexane in b).

- e) **Filtering material** In the case of using filter paper, that of circular or cylindrical and of silica fibre made among the filtering materials specified in **JIS K 0901** is used. In the case of dust tube, that of glass fibre or silica fibre made is used. For any one, prior to the test ultrasonic cleaning is carried out by using acetone and toluene respectively for 30 min and vacuum drying is carried out. Or the filtering material of silica fibre is heated at 600 °C by using electric furnace approximately for 6 h. The washed or heated filtering material, after confirming that the blank test component of dioxins and other component to hinder analysis are not contained, shall be used.
- f) **Adsorbent** Styrene-divinylbenzene copolymer. This adsorbent is washed with acetone prior to use and the washing is carried out by means of Soxhlet extraction with toluene for 16 h or longer. Or, ultrasonic cleaning is carried out by using in order of acetone, toluene (twice) for 30 min respectively. Then, it is heated at 50 °C or lower in vacuum drier for 8 h and preserved in a hermetical container. The washed adsorbent is carried out by the similar operation to the extraction from the exhaust gas sample, the extracted solution is concentrated and used after it is confirmed that the blank test component of dioxins and other component to hinder analysis are not contained.

Information : "XAD-2" is generally available as adsorbent, however that is not the case of recommending this. If at least the equivalent effect is proved to obtain, the other one may be used.

3.4 Precautions on operation

- a) For the filter-collecting part, the moisture in exhaust gas shall be made so as to be in the range between the temperature without condensation and 120 °C. When the temperature of exhaust gas is high, it shall be made so as not to exceed 120 °C by adjusting the length of connecting pipe to sampling pipe part, and so on. When the temperature of exhaust gas is low, it shall be required to warm at the degree not to condense the moisture.
- b) When the isokinetic sampling has become difficult caused by the increase of resistance in filtering material during the isokinetic sampling, stop the suction for a time and exchange the filtering material in filter-collecting part.
- c) Put the liquid-collecting part in ice bath or dry ice bath so as to maintain each absorption bottle at 5 °C to 6 °C or lower.
- d) Maintain the adsorption-collecting part at 30 °C or lower. When the ambient temperature is high, cool it in ice bath, etc. as similar to the liquid-collecting part.
- e) Shade all of the collecting-parts from light.

4 JIS type II apparatus

4.1 Scope in use This apparatus is applicable to the sample gas in the following range. For the use in the range other than this range, it shall be required that the confirmation specified in 5.2 of the text is carried out.

- a) **Objective exhaust gas** The exhaust gas from the incineration furnace for general and industrial waste.

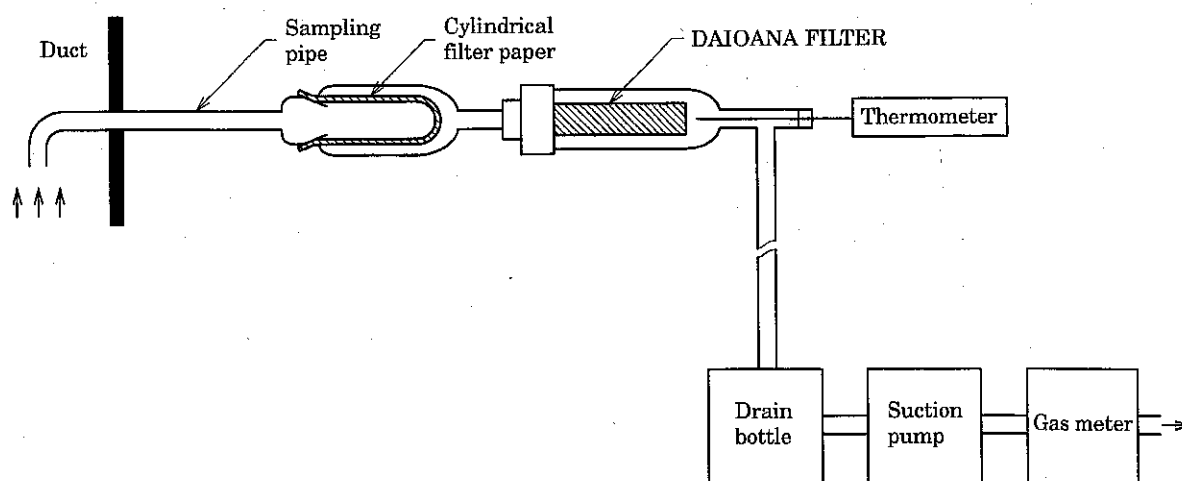
- b) **Moisture content** Not more than 36 %.
- c) **Concentration of carbon monoxide** Not more than 670 ppm in average actual measured concentration.

4.2 Construction of apparatus This apparatus shall be composed of sampling pipe part followed by filter-collecting part and adsorption-collecting part. The outline of the construction of this apparatus is as shown in annex 1 figure 2.

- a) **Filter-collecting part** For the filter-collecting part, the type II dust collecting device specified in 8.3 of JIS Z 8808 shall be used. In the case of using filter paper, the circular or cylindrical one made of silica fibre shall be used. In the case of dust tube, it shall be packed with glass fibre or silica fibre. In any case, it shall be confirmed that blank test component and other interfering component are not found in presence prior to the use.

When the dust content is little and the sampling and measurement are not hindered, the filter-collecting part may be omitted. And, depending on the kind of combustion equipment, the dust tube containing silica fibre, etc. shall be used in front of cylindrical filter paper.

- b) **Adsorption-collecting part** The adsorption-collecting shall be carried out by using a cylindrical filter type adsorbent of alumina adsorbent specially processed and moulded. For ensuring of collecting efficiency at this adsorption-collecting part, that shall be used at not more than 33 L/min (wet gas, 120 °C, 101.32 kPa) in passing flow rate of collecting-part and not exceeding 3.8 m³ (dry gas, 0 °C, 101.32 kPa) in sampling amount of sample gas.
- c) **Connecting-part** The connecting pipe from filter-collecting part to liquid-collecting part shall be made as short as possible and that of glass or fluorocarbon resin shall be used. For the connection for each part, an interchangeable ball and socket joint pipe or fluorocarbon resin pipe joint, etc. shall be used and grease shall not be used at the connecting part.



Annex 1 Figure 2 Schematic diagram of construction of JIS type II sampling apparatus

4.3 Reagents The reagents to be used for this apparatus shall be as follows. It shall be confirmed by blank test, etc. that these reagents do not hinder the measurement.

- a) **Filtering material** In the case of using filter paper, that of circular or cylindrical and silica fibre made among the filtering materials specified in **JIS K 0901** is used. In the case of using dust tube, that of glass fibre or silica fibre made is used. For any one, prior to the use ultrasonic cleaning is carried out by using acetone and toluene respectively for 30 min and vacuum drying is carried out. Or the filtering material of silica fibre is heated at 600 °C by using electric furnace approximately for 6 h. The washed and heated filtering material, after confirming that the blank test component of dioxins and other component to hinder analysis are not contained, shall be used.
- b) **Adsorbent** Cylindrical filter type adsorbent of alumina adsorbent specially processed and moulded. This adsorbent can be usually available in washed condition, therefore the washing prior to use needs not especially to carry out. Provided that it shall be used after confirming every lot that dioxins and other components to hinder analysis are not contained.

Information : This adsorbent is on the market as the trade name of "DAIOANA FILTER".

4.4 Precaution on operation

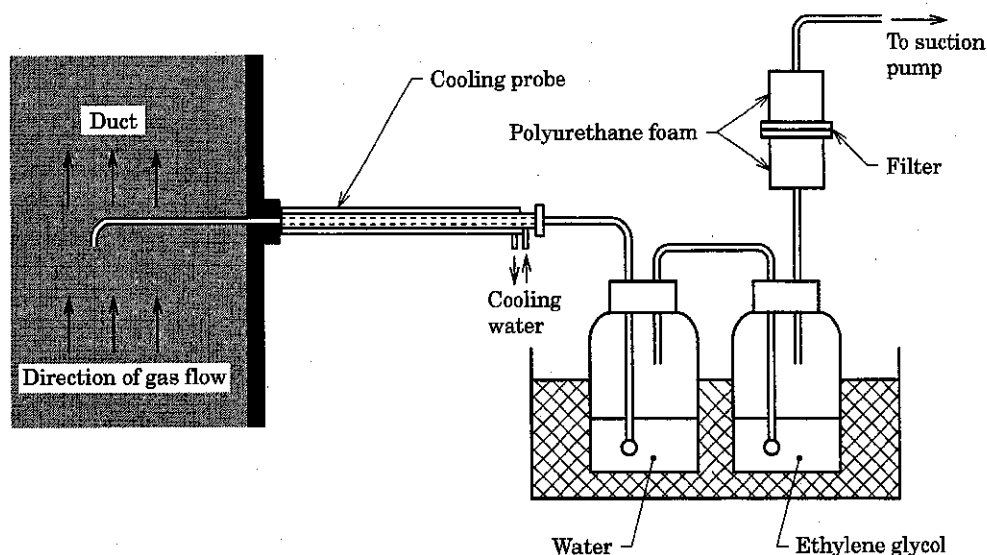
- a) For the filter-collecting part and adsorption-collecting part, the moisture in exhaust gas shall be made so as to be in the range between the temperature without condensation and 120 °C. When the temperature of exhaust gas is high, it shall be made so as not to exceed 120 °C by adjusting the length of connecting pipe to sampling pipe part, and so on. When the temperature of exhaust gas is low, it shall be required to warm at the degree not to condense the moisture.
- b) When the isokinetic sampling has become difficult caused by the increase of resistance in filtering material during isokinetic sampling, stop the suction for a time and exchange the filtering material in collecting part.
- c) Shade all of the collecting parts from light.
- d) For ensuring of collecting efficiency, use at not more than 33 L/min (wet gas, 120 °C, 101.32 kPa) in passing flow rate of collecting-part and not exceeding 3.8 m³ (dry gas, 0 °C, 101.32 kPa) in sampling amount of sample gas.
- e) When the moisture condenses in front of adsorption-collecting part in the case of using a cooling probe, etc., avoid the use. And also when the tar content exceeding 3 mg/m³ (0 °C, 101.32 kPa) is found in presence, do not use.

5 JIS type III apparatus

5.1 Scope in use This apparatus can be applicable to almost exhaust gas. Provided that the exhaust gas having very high concentration of dust and very high concentration of dioxins shall be used after carrying out the confirmation specified in 5.2 of the text.

5.2 Construction of apparatus This apparatus shall be composed of sampling pipe part followed by cooling probe, liquid-collecting part and adsorption- and filter-collecting part. The outline of the construction of this apparatus is as shown in annex 1 figure 3.

- a) **Cooling probe** That which cools the sample gas so as to be 20 °C or lower at the liquid-collecting part in the rear part.
- b) **Liquid-collecting part** For the liquid-collecting part, water is put in the first stage and 1,2-ethanediol (ethylene glycol) is put in the second stage, and the sample gas is cooled to 20 °C or lower at the first stage to condense the moisture in the gas and dust is collected together. At the second stage, the dust and gas condition of dioxins passed through the first stage are collected. The cooling in water bath, etc. shall be required.
- c) **Adsorption- and filter-collecting part** That which the adsorbent such as urethane foam, etc. is connected before and behind, and the gas condition of dioxins and small particle size of dust passed through the liquid-collecting part are collected.
- d) **Connecting part** The connecting pipe from filter-collecting part to liquid-collecting part shall be made as short as possible and that of glass or fluorocarbon resin shall be used. For the connection for each part, an interchangeable ball and socket joint pipe or fluorocarbon resin pipe joint, etc. shall be used and grease shall not be used at the connecting part.



Annex 1 Figure 3 Schematic diagram of construction of JIS type III apparatus

5.3 Reagents The reagents to be used for this apparatus shall be as follows. It shall be confirmed by blank test, etc. that these reagents as not hinder the measurement.

- a) **Water** Water of grade A4 (or A3) specified in **JIS K 0557**.

- b) **1,2-Ethanediol** Ethylene glycol specified in **JIS K 8105** or that of equivalent quality.
- c) **Filter** For the filter, that of silica fibre or glass fibre shall be used. Prior to the test, ultrasonic cleaning is carried out by using acetone specified in **JIS K 8040** and toluene specified in **JIS K 8680** respectively for 30 min and vacuum drying is carried out. Or, the filtering material of silica fibre is heated at 600 °C by using electric furnace approximately for 6 h. The washed and heated filtering material, after confirming that the blank test component of dioxins and other component to hinder analysis are not contained, shall be used.
- d) **Adsorbent** Polyurethane foam or that of equivalent performance to this. This adsorbent, prior to use, is washed with water and acetone specified in **JIS K 8040** and the washing is carried out by means of Soxhlet extraction with acetone for 16 h to 24 h. Or, ultrasonic cleaning is carried out with using acetone thrice for 30 min. Then, it is heated at 50 °C or lower in vacuum drier for 8 h, dried sufficiently and preserved in hermetical container. The washed adsorbent is carried out by the similar operation to the extraction from the exhaust gas sample, the extracted solution is concentrated and used after it is confirmed that the blank test component of dioxins and other component to hinder analysis are not contained.

5.4 Precautions on operation

- a) The cooling probe cools so that the sample gas becomes 20 °C or lower at the rear stage of liquid-collecting part.
- b) Confirm that the sample gas can be cooled up to 20 °C at the first stage of liquid-collecting part.
- c) Shade all of the collecting part from light.

Annex 2 (informative)

Example for use of internal standard substance

Introduction This annex describes the example for use of internal standard substance and does not constitute part of provisions.

For the use of internal standard substance, the use method is partially different depending on the analytical conditions of GC/MS, therefore the detail of use is not especially specified and only the lowest limit of requirements are specified. The general examples of use are shown herein and the standard solution for preparation of calibration curve in such a case, the objective substance for analysis at determination and the internal standard substance corresponding to this, etc. are shown as example. It is required that the use is examined at each laboratory according to the operation procedure, the analytical conditions of GC/MS as referring to this.

1 Example for use of internal standard substance

Internal standard substance	Example 1			Example 2		
	Sampling spike	Cleanup spike	Syringe spike	Sampling spike	Cleanup spike	Syringe spike
¹³ C ₁₂ -1, 2, 7, 8-TeCDF				○		
¹³ C ₁₂ -2, 3, 7, 8-TeCDF		○			○	
¹³ C ₁₂ -1, 2, 3, 4-TeCDD			○			○
¹³ C ₁₂ -2, 3, 7, 8-TeCDD		○			○	
³⁷ Cl ₄ -2, 3, 7, 8-TeCDD						
¹³ C ₁₂ -1, 2, 3, 7, 8-PeCDF	○				○	
¹³ C ₁₂ -2, 3, 4, 7, 8-PeCDF		○				
¹³ C ₁₂ -1, 2, 3, 7, 8-PeCDD		○			○	
¹³ C ₁₂ -1, 2, 3, 4, 7, 8-HxCDF		○				
¹³ C ₁₂ -1, 2, 3, 6, 7, 8-HxCDF		○			○	
¹³ C ₁₂ -1, 2, 3, 7, 8, 9-HxCDF	○					
¹³ C ₁₂ -2, 3, 4, 6, 7, 8-HxCDF		○				
¹³ C ₁₂ -1, 2, 3, 4, 7, 8-HxCDD		○				
¹³ C ₁₂ -1, 2, 3, 6, 7, 8-HxCDD		○			○	
¹³ C ₁₂ -1, 2, 3, 7, 8, 9-HxCDD			○			○
¹³ C ₁₂ -1, 2, 3, 4, 6, 7, 8-HpCDF		○			○	
¹³ C ₁₂ -1, 2, 3, 4, 7, 8, 9-HpCDF	○			○		
¹³ C ₁₂ -1, 2, 3, 4, 6, 7, 8-HpCDD		○			○	
¹³ C ₁₂ -1, 2, 3, 4, 6, 7, 8, 9-OCDF		○				
¹³ C ₁₂ -1, 2, 3, 4, 6, 7, 8, 9-OCDD		○			○	
¹³ C ₁₂ -2, 3', 4', 5-TeCB (#70)				○		
¹³ C ₁₂ -3, 3', 4, 4'-TeCB (#77)		○			○	
¹³ C ₁₂ -3, 4, 4', 5-TeCB (#81)		○			○	
¹³ C ₁₂ -3, 3', 4, 4', 5-PeCB (#126)		○			○	
¹³ C ₁₂ -2, 3, 3', 4, 4'-PeCB (#105)			○			
¹³ C ₁₂ -2, 3, 4, 4', 5-PeCB (#114:)						○
¹³ C ₁₂ -2, 3', 4, 4', 5-PeCB (#118)		○			○	
¹³ C ₁₂ -2', 3, 4, 4', 5-PeCB (#123)						
¹³ C ₁₂ -2, 3, 3', 4, 4', 5-HxCB (#156)		○			○	
¹³ C ₁₂ -2, 3, 3', 4, 4', 5'-HxCB (#157)						
¹³ C ₁₂ -2, 3', 4, 4', 5, 5'-HxCB (#167)		○				
¹³ C ₁₂ -3, 3', 4, 4', 5, 5'-HxCB (#169)		○			○	
¹³ C ₁₂ -2, 3, 3', 4, 4', 5, 5'-HpCB (#189)		○			○	
¹³ C ₁₂ -2, 2', 3, 3', 4, 4', 5-HpCB (#170)						
¹³ C ₁₂ -2, 2', 3, 3', 4, 4', 5, 5'-HpCB (#180)						

Remarks : The numerals in parentheses indicate IUPAC No.

2 Example for preparation of reference solution for preparation of calibration curve (case of example 1 in 1)

Standard substance	Concentration (ng/ml)				
	STD1	STD2	STD3	STD4	STD5
2, 3, 7, 8-TeCDD	0.4	2.0	10	40	200
1, 2, 3, 7, 8-PeCDD					
1, 2, 3, 4, 7, 8-HxCDD	1.0	5.0	25	100	500
1, 2, 3, 6, 7, 8-HxCDD					
1, 2, 3, 7, 8, 9-HxCDD					
1, 2, 3, 4, 6, 7, 8-HpCDD					
OCDD					
2, 3, 7, 8-TeCDF	0.4	2.0	10	40	200
1, 2, 3, 7, 8-PeCDF					
2, 3, 4, 7, 8-PeCDF					
1, 2, 3, 4, 7, 8-HxCDF	1.0	5.0	25	100	500
1, 2, 3, 6, 7, 8-HxCDF					
1, 2, 3, 7, 8, 9-HxCDF					
2, 3, 4, 6, 7, 8-HxCDF					
1, 2, 3, 4, 6, 7, 8-HpCDF					
1, 2, 3, 4, 7, 8, 9-HpCDF					
OCDF	2.0	10	50	200	1 000
¹³ C ₁₂ -2, 3, 7, 8-TeCDD	100	100	100	100	100
¹³ C ₁₂ -1, 2, 3, 4-TeCDD					
¹³ C ₁₂ -1, 2, 3, 7, 8-PeCDD					
¹³ C ₁₂ -1, 2, 3, 4, 7, 8-HxCDD					
¹³ C ₁₂ -1, 2, 3, 6, 7, 8-HxCDD					
¹³ C ₁₂ -1, 2, 3, 7, 8, 9-HxCDD					
¹³ C ₁₂ -1, 2, 3, 4, 7, 8, 9-HpCDD	200	200	200	200	200
¹³ C ₁₂ -OCDD					
¹³ C ₁₂ -2, 3, 7, 8-TeCDF	100	100	100	100	100
¹³ C ₁₂ -1, 2, 3, 7, 8-PeCDF					
¹³ C ₁₂ -2, 3, 4, 7, 8-PeCDF					
¹³ C ₁₂ -1, 2, 3, 4, 7, 8-HxCDF					
¹³ C ₁₂ -1, 2, 3, 6, 7, 8-HxCDF					
¹³ C ₁₂ -1, 2, 3, 7, 8, 9-HxCDF					
¹³ C ₁₂ -2, 3, 4, 6, 7, 8-HxCDF					
¹³ C ₁₂ -1, 2, 3, 4, 6, 7, 8-HpCDF					
¹³ C ₁₂ -1, 2, 3, 4, 7, 8, 9-HpCDF					
¹³ C ₁₂ -OCDF	200	200	200	200	200
3, 3', 4, 4'-TeCB (#77)	1.0	5.0	25	100	500
3, 4, 4', 5-TeCB (#81)					
2, 3, 3', 4, 4'-PeCB (#105)					
2, 3, 4, 4', 5-PeCB (#114)					
2, 3', 4, 4', 5-PeCB (#118)					
2', 3, 4, 4', 5-PeCB (#123)					
3, 3', 4, 4', 5-PeCB (#126)					
2, 3, 3', 4, 4', 5-HxCB (#156)					
2, 3, 3', 4, 4', 5'-HxCB (#157)					
2, 3', 4, 4', 5, 5'-HxCB (#167)					
3, 3', 4, 4', 5, 5'-HxCB (#169)					

2 (concluded)

Standard substance	Concentration (ng/ml)				
	STD1	STD2	STD3	STD4	STD5
2, 2', 3, 3', 4, 4', 5-HpCB (#170)					
2, 2', 3, 4, 4', 5, 5'-HpCB (#180)					
2, 3, 3', 4, 4', 5, 5'-HpCB (#189)					
¹³ C ₁₂ -3, 3', 4, 4'-TeCB (#77)					
¹³ C ₁₂ -3, 4, 4', 5-TeCB (#81)					
¹³ C ₁₂ -2, 3, 3', 4, 4'-PeCB (#105)					
¹³ C ₁₂ -2, 3', 4, 4', 5-PeCB (#118)					
¹³ C ₁₂ -3, 3', 4, 4', 5-PeCB (#126)	100	100	100	100	100
¹³ C ₁₂ -2, 3, 3', 4, 4', 5-HxCB (#156)					
¹³ C ₁₂ -2, 3', 4, 4', 5, 5'-HxCB (#167)					
¹³ C ₁₂ -3, 3', 4, 4', 5, 5'-HxCB (#169)					
¹³ C ₁₂ -2, 3, 3', 4, 4', 5, 5'-HpCB (#189)					

3 Example for correspondence of objective substance for measurement, standard substance to cleanup spike internal standard substance (case of example 1 in 1)

Standard substance	Corresponding cleanup spike internal standard substance
2, 3, 7, 8-TeCDF	$^{13}\text{C}_{12}$ -2, 3, 7, 8-TeCDF
2, 3, 7, 8-TeCDD	$^{13}\text{C}_{12}$ -2, 3, 7, 8-TeCDD
1, 2, 3, 7, 8-PeCDF	$^{13}\text{C}_{12}$ -2, 3, 4, 7, 8-PeCDF
2, 3, 4, 7, 8-PeCDF	
1, 2, 3, 7, 8-PeCDD	$^{13}\text{C}_{12}$ -1, 2, 3, 7, 8-PeCDD
1, 2, 3, 4, 7, 8-HxCDF	$^{13}\text{C}_{12}$ -1, 2, 3, 4, 7, 8-HxCDF
1, 2, 3, 6, 7, 8-HxCDF	
1, 2, 3, 7, 8, 9-HxCDF	$^{13}\text{C}_{12}$ -1, 2, 3, 6, 7, 8-HxCDF
2, 3, 4, 6, 7, 8-HxCDF	
1, 2, 3, 4, 7, 8-HxCDD	$^{13}\text{C}_{12}$ -2, 3, 4, 6, 7, 8-HxCDF
1, 2, 3, 6, 7, 8-HxCDD	
1, 2, 3, 4, 7, 8-HxCDD	$^{13}\text{C}_{12}$ -1, 2, 3, 4, 7, 8-HxCDD
1, 2, 3, 6, 7, 8-HxCDD	
1, 2, 3, 7, 8, 9-HxCDD	$^{13}\text{C}_{12}$ -1, 2, 3, 6, 7, 8-HxCDD
1, 2, 3, 4, 6, 7, 8-HpCDF	
1, 2, 3, 4, 7, 8, 9-HpCDF	$^{13}\text{C}_{12}$ -1, 2, 3, 4, 6, 7, 8-HpCDF
1, 2, 3, 4, 6, 7, 8-HpCDD	
1, 2, 3, 4, 6, 7, 8, 9-OCDF	$^{13}\text{C}_{12}$ -1, 2, 3, 4, 6, 7, 8-HpCDD
1, 2, 3, 4, 6, 7, 8, 9-OCDD	
3, 3', 4, 4'-TeCB (#77)	$^{13}\text{C}_{12}$ -1, 2, 3, 4, 6, 7, 8, 9-OCDF
3, 4, 4', 5-TeCB (#81)	$^{13}\text{C}_{12}$ -1, 2, 3, 4, 6, 7, 8, 9-OCDD
2, 3, 3', 4, 4'-PeCB (#105)	$^{13}\text{C}_{12}$ -3, 3', 4, 4'-TeCB (#77)
2, 3, 4, 4', 5-PeCB (#114)	
2, 3', 4, 4', 5-PeCB (#118)	
2', 3, 4, 4', 5-PeCB (#123)	
3, 3', 4, 4', 5-PeCB (#126)	$^{13}\text{C}_{12}$ -3, 4, 4', 5-TeCB (#81)
2, 3, 3', 4, 4', 5-HxCB (#156)	$^{13}\text{C}_{12}$ -2, 3', 4, 4', 5-PeCB (#118)
2, 3, 3', 4, 4', 5'-HxCB (#157)	
2, 3', 4, 4', 5, 5'-HxCB (#167)	
3, 3', 4, 4', 5, 5'-HxCB (#169)	
2, 2', 3, 3', 4, 4', 5-HpCB (#170)	$^{13}\text{C}_{12}$ -3, 3', 4, 4', 5-PeCB (#126)
2, 2', 3, 4, 4', 5, 5'-HpCB (#180)	
2, 3, 3', 4, 4', 5, 5'-HpCB (#189)	

4 Example for correspondence of cleanup spike internal standard substance to syringe spike internal standard substance (case of example 1 in 1)

Cleanup spike internal standard substance	Corresponding syringe internal standard substance
$^{13}\text{C}_{12-2, 3, 7, 8}\text{-TeCDF}$	$^{13}\text{C}_{12-1, 2, 3, 4}\text{-TeCDD}$
$^{13}\text{C}_{12-2, 3, 7, 8}\text{-TeCDD}$	
$^{13}\text{C}_{12-2, 3, 4, 7, 8}\text{-PeCDF}$	
$^{13}\text{C}_{12-1, 2, 3, 7, 8}\text{-PeCDD}$	
$^{13}\text{C}_{12-1, 2, 3, 4, 7, 8}\text{-HxCDF}$	$^{13}\text{C}_{12-1, 2, 3, 7, 8, 9}\text{-HxCDD}$
$^{13}\text{C}_{12-1, 2, 3, 6, 7, 8}\text{-HxCDF}$	
$^{13}\text{C}_{12-2, 3, 4, 6, 7, 8}\text{-HxCDF}$	
$^{13}\text{C}_{12-1, 2, 3, 4, 7, 8}\text{-HxCDD}$	
$^{13}\text{C}_{12-1, 2, 3, 6, 7, 8}\text{-HxCDD}$	
$^{13}\text{C}_{12-1, 2, 3, 4, 6, 7, 8}\text{-HpCDF}$	
$^{13}\text{C}_{12-1, 2, 3, 4, 6, 7, 8}\text{-HpCDD}$	
$^{13}\text{C}_{12-1, 2, 3, 4, 6, 7, 8, 9}\text{-OCDF}$	
$^{13}\text{C}_{12-1, 2, 3, 4, 6, 7, 8, 9}\text{-OCDD}$	$^{13}\text{C}_{12-2, 3, 3', 4, 4'}\text{-PeCB (#105)}$
$^{13}\text{C}_{12-3, 3', 4, 4'}\text{-TeCB (#77)}$	
$^{13}\text{C}_{12-3, 4, 4', 5}\text{-TeCB (#81)}$	
$^{13}\text{C}_{12-2, 3', 4, 4', 5}\text{-PeCB (#118)}$	
$^{13}\text{C}_{12-3, 3', 4, 4', 5}\text{-PeCB (#126)}$	
$^{13}\text{C}_{12-2, 3, 3', 4, 4', 5}\text{-HxCB (#156)}$	
$^{13}\text{C}_{12-2, 3', 4, 4', 5, 5'}\text{-HxCB (#167)}$	
$^{13}\text{C}_{12-3, 3', 4, 4', 5, 5'}\text{-HxCB (#169)}$	
$^{13}\text{C}_{12-2, 3, 3', 4, 4', 5, 5'}\text{-HpCB (#189)}$	

5 Example for correspondence of sampling spike internal standard substance to cleanup spike internal standard substance (case of example 1 in 1)

Sampling spike internal standard substance	Corresponding cleanup spike internal standard substance
$^{13}\text{C}_{12-1, 2, 3, 7, 8}\text{-PeCDF}$	$^{13}\text{C}_{12-2, 3, 4, 7, 8}\text{-PeCDF}$
$^{13}\text{C}_{12-1, 2, 3, 7, 8, 9}\text{-HxCDF}$	$^{13}\text{C}_{12-1, 2, 3, 6, 7, 8}\text{-HxCDF}$
$^{13}\text{C}_{12-1, 2, 3, 4, 7, 8, 9}\text{-HpCDF}$	$^{13}\text{C}_{12-1, 2, 3, 4, 6, 7, 8}\text{-HpCDF}$

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